Group 11

Smart Bus Booking & Operations Management System

Version 1.8

Group 11 — PA3 Use-Case Specifications, edition 1.8. Inherits requirements baseline 1.5 plus QĐ-07 (08/11/2026): Gemini Developer API replaces the earlier AI provider. Leader approved PA3 design; 20 goals + C01, 96 primary BRs and 62 acceptance designs retained. Software execution is not certified.

Source: PA2 – Group 11 | Team Work Plan & Shared UC Template, consolidated specifications 08A–08C, revised with the project lead's approved requirements decisions on 23 October 2026. Business policies are defined in this document; the configurable AI model is assigned to PA3 architecture.

Revision History

Date	Version	Description	Author
23/Oct/26	1.0	Completed 20 use-case specifications from PA2; unresolved questions retained.	Group 11
23/Oct/26	1.1	Applied approved decisions across all 20 UCs; aligned per-seat Tickets, lifecycle states, refunds and operational scope.	Group 11
24/Oct/26	1.2	Quality review: completed alternative flows, aligned entry paths/boarding checks and replaced overview with a larger-text layout.	Liên Trung Hiếu
04/Nov/26	1.3	Feedback correction: explicit temporary P08; external AI actor; C01 inclusion and P10/P06 extension; read-only A04 naming; BR identifiers; editable model and current navigation. Historical policies retained.	Team 11 / lead-approved revision
04/Nov/26	1.4	Second quality pass: note-free UML views and explicit extension-point notation; complete UC/BR/NFR/design/test traceability, access matrix and 62 designed acceptance cases. DEC-01/02 pending lead decisions; no executed-test or member-sign-off claim.	Team 11 / editorial quality pass
04/Nov/26	1.5	Lead approved DEC-01 A and DEC-02 A: selected boarding date and whole-Booking integer-VND half-up refund. Added BR-P04-05 / BR-P10-07, finalized TC-09/25 and synchronized all date/refund references. No software test execution is claimed. Layout update on 4 November: lead-approved top-actor / common vertical association layout applied to all six model views (model 1.5); relationships and requirements unchanged.	Liên Trung Hiếu / approved decisions
06/Nov/26	1.6	PA3 edition inherits requirements baseline 1.5. Made existing Vietnamese-only alternatives explicit in P11/P12; synchronized all six model images and editable sources to visual edition 2.0; retained 20 goals, C01, 96 BRs and 62 acceptance designs. No executed-test/member-sign-off claim.	Team 11 / assisted consolidation

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06/Nov/26	1.7	Editorial integration: six diagrams/references synchronized to visual edition 2.1; P08 A2 explicitly rejects repeated seat identifiers. Baseline 1.5, 96 BRs/62 acceptance designs retained; final review pending.	Group 11 / editorial integration
08/Nov/26	1.8	QĐ-07: leader approved Gemini Developer API Free Tier, replacing the earlier provider in BR-P11-05/P12-06. Approved FAQ/contact handling and final PA3 design. Preserved UC/BR IDs, 20 goals+C01, 62 acceptance designs and generic model 2.1. No software-test or member-sign-off claim.	Liên Trung Hiếu / approved decision

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Use case Specification Document

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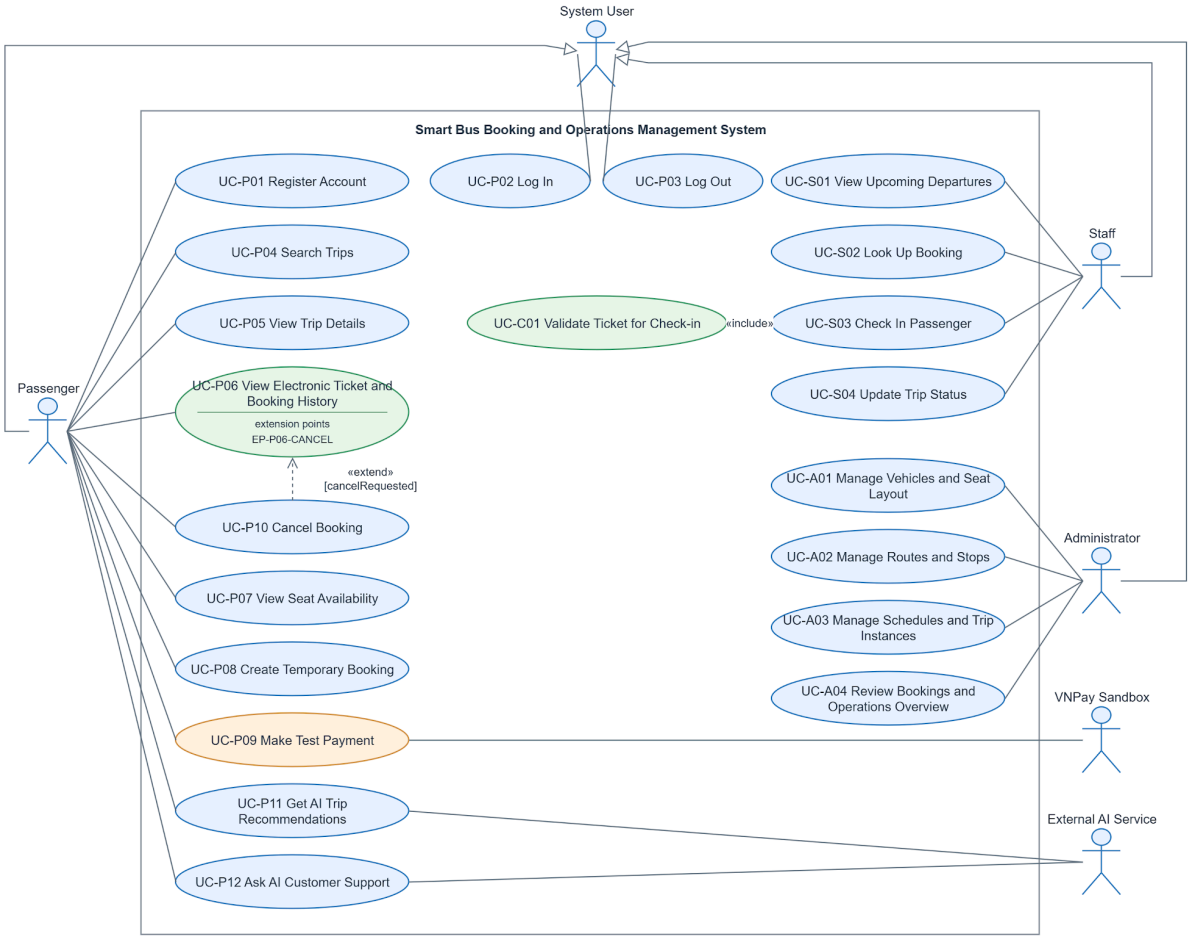
UC-A04 Review Bookings and Operations Overview

Supporting behavior: UC-C01 — Appendix A

BR reference convention — Appendix B

Use-Case Model — Overview (visual edition 2.1; requirements baseline 1.5). Five detailed views are in Appendix C. Editable source: PA3_Group11_UseCaseModel_v2.1.drawio [REF-01]. Full-size views: the same six PNG/SVG exports listed in Appendix D [REF-02]. Relationships and business policies are unchanged.

Use-Case Model — Overview



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This document specifies the behavior of 12 Passenger, 4 Staff and 4 Administrator use cases using the supplied RUP template. UC-P02 and UC-P03 are shared by all three roles.

Requirements baseline 1.5 — Feedback corrections approved on 4 November 2026 preserve the October business policies: P08 creates a temporary Pending Booking; P09 remains separate. External AI Service supports P11/P12; QĐ-07 on 8 November 2026 replaces the earlier provider with Google Gemini Developer API. This is a later approved amendment. S03 includes C01; P10 extends P06 only at EP-P06-CANCEL. Numbered BR identifiers below are the authoritative policy references for PA3 design and tests. Acceptance designs and decision gates are in Appendix D — Acceptance & Traceability. On 4 November 2026 the project lead approved DEC-01 option A (selected boarding date) and DEC-02 option A (whole-VND half-up rounding); the authoritative rules are BR-P04-05 and BR-P10-07.

Entry convention — Preconditions describe normal invocation. Authentication/authorization alternatives also cover invalid direct entry or session expiry; no protected data or modifying operation is permitted until the corresponding checks pass. Eligibility is rechecked immediately before a protected write.

1. UC-P01 Register Account

1.1 Register Account

1.1.1 Brief Description

Create a passenger account so the Passenger can use account-based functions.

Related Feature(s): F01

Primary Actor: Passenger

Supporting Actor(s): None

Trigger: The Passenger chooses to register a new account.

1.2 Flow of Events

1.2.1 Basic Flow

1. Passenger opens registration.

2. The system displays fields for full name, email, telephone number, password and password confirmation.

3. Passenger completes all required fields and submits the form.

4. The system validates required values, matching passwords and email uniqueness.

5. The system creates an active Passenger account with email as the login identifier.

6. The system confirms registration and directs Passenger to Log In (UC-P02), without starting a session.

1.2.2 Alternative Flows

1.2.2.1 A1 — Invalid or incomplete registration

A1.1 At Step 4, required information is missing or invalid, or password confirmation does not match.

A1.2 The system identifies the fields to correct and returns to Step 3; no account is created.

1.2.2.2 A2 — Email already registered

A2.1 At Step 4, the email is already used by an account.

A2.2 The system rejects the duplicate and offers correction or UC-P02; no existing account is changed.

1.2.2.3 A3 — Account cannot be saved

A3.1 At Step 5, the account cannot be saved.

A3.2 The system reports failure without claiming success; no partial account is usable.

1.3 Special Requirements

1.3.1 Business Rules

[BR-P01-01] Full name, email, telephone number, password and password confirmation are mandatory. Email is the unique login identifier; telephone number is the journey contact.

[BR-P01-02] Public registration creates Passenger accounts only. Staff and Administrator accounts are provisioned before use, outside this use case.

[BR-P01-03] Successful registration does not automatically log the Passenger in. Passwords are not displayed or stored as plain text.

1.3.2 Approved Decisions and Scope

Email-based registration and explicit login after registration are approved. Account provisioning and account-management screens for Staff/Administrator are outside the 20-use-case scope.

Specification owner: Nguyễn Khánh Hoàng | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

1.4 Preconditions

1.4.1 Required System and Actor State

The Passenger has no active authenticated session. Public registration is available.

1.5 Postconditions

1.5.1 Success

One new Passenger account exists, and Passenger can continue to UC-P02.

1.5.2 Failure and Minimal Guarantees

No duplicate or partially usable account is created. Existing accounts remain unchanged.

1.6 Extension Points

1.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

2. UC-P02 Log In

2.1 Log In

2.1.1 Brief Description

Authenticate and start a session so the user can access functions that require an account.

Related Feature(s): F01

Primary Actors: Passenger, Staff, Administrator

Supporting Actor(s): None

Trigger: The User chooses to log in.

2.2 Flow of Events

2.2.1 Basic Flow

1. The User opens Log In.

2. The system displays email and password inputs.

3. The User submits email and password.

4. The system validates the inputs and checks whether the login attempt is temporarily restricted.

5. The system verifies the credentials and checks that the account is active.

6. The system starts an authenticated session, resets the consecutive-failure counter and determines the assigned role.

7. The system displays only functions authorized for that role.

2.2.2 Alternative Flows

2.2.2.1 A1 — Incomplete credentials

A1.1 At Step 4, an input is missing or invalid.

A1.2 The system requests correction and returns to Step 3; no session is created.

2.2.2.2 A2 — Incorrect credentials

A2.1 At Step 5, the credentials do not match.

A2.2 The system gives a generic login-failure message and records the failed attempt. The User may retry at Step 3 subject to the temporary restriction.

2.2.2.3 A3 — Temporary login restriction

A3.1 At Step 4, five consecutive failed attempts have occurred within 15 minutes.

A3.2 The system temporarily restricts further login attempts for five minutes and communicates the retry condition. No authenticated session is started and the account is not permanently locked.

2.2.2.4 A4 — Inactive account

A4.1 At Step 5, the account is disabled or inactive.

A4.2 The system denies login without opening a session; account administration is outside this use case.

2.3 Special Requirements

2.3.1 Business Rules

[BR-P02-01] Passenger, Staff and Administrator share UC-P02; access after login follows the assigned role.

[BR-P02-02] Five consecutive incorrect attempts within 15 minutes trigger a five-minute temporary restriction; successful authentication resets the consecutive-failure counter.

[BR-P02-03] Login restrictions must not become a permanent account lock. Disabled accounts cannot log in.

[BR-P02-04] Passenger sessions expire after 30 minutes of inactivity; Staff and Administrator sessions expire after 15 minutes. Protected requests require an active session.

2.3.2 Approved Decisions and Scope

Use email and password. Temporary failed-login restriction, inactive-account rejection and role-specific inactivity timeouts are approved.

Specification owner: Nguyễn Khánh Hoàng | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

2.4 Preconditions

2.4.1 Required System and Actor State

The User has no active authenticated session. The login function is available to Passenger, Staff and Administrator.

2.5 Postconditions

2.5.1 Success

An active account has an authenticated session and role-appropriate access.

2.5.2 Failure and Minimal Guarantees

No authenticated session is created on failure. Credentials and protected records are not disclosed.

2.6 Extension Points

2.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

3. UC-P03 Log Out

3.1 Log Out

3.1.1 Brief Description

End the current session securely.

Related Feature(s): F01

Primary Actors: Passenger, Staff, Administrator

Supporting Actor(s): None

Trigger: The User (Passenger, Staff or Administrator) chooses to log out.

3.2 Flow of Events

3.2.1 Basic Flow

1. The User selects Log Out.

2. The system invalidates the active session and removes access to protected functions.

3. The system clears protected information displayed for the session and returns the User to the public view.

3.2.2 Alternative Flows

3.2.2.1 A1 — Session already expired

A1.1 At Step 2, the session is already inactive.

A1.2 The system keeps protected access closed, clears protected display data and offers UC-P02; the use case ends.

3.2.2.2 A2 — Logout cannot be confirmed

A2.1 At Step 2, the system cannot confirm successful logout.

A2.2 The system reports the problem without claiming success and does not alter account, booking or payment data.

3.3 Special Requirements

3.3.1 Business Rules

[BR-P03-01] All three roles share this logout use case.

[BR-P03-02] Passenger inactivity timeout is 30 minutes; Staff/Administrator timeout is 15 minutes. An expired session cannot perform protected actions.

[BR-P03-03] Logout does not cancel a Booking or extend, release or recreate a Seat Hold; the hold follows

its original expiry.

3.3.2 Approved Decisions and Scope

Role-specific inactivity timeouts are approved. Continued booking actions after expiry require UC-P02 and a fresh check of ownership, Booking status and hold validity.

Specification owner: Nguyễn Khánh Hoàng | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

3.4 Preconditions

3.4.1 Required System and Actor State

The User has previously logged in; the current session may be active or may have expired.

3.5 Postconditions

3.5.1 Success

The current session is invalidated and protected information is no longer available through it.

3.5.2 Failure and Minimal Guarantees

Account and journey records remain unchanged; expired sessions are never restored by logout.

3.6 Extension Points

3.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

4. UC-P04 Search Trips

4.1 Search Trips

4.1.1 Brief Description

Find available departures matching the chosen route and travel date.

Related Feature(s): F02

Primary Actor: Passenger

Supporting Actor(s): None

Trigger: The Passenger submits trip search criteria.

4.2 Flow of Events

4.2.1 Basic Flow

1. Passenger enters origin Stop, destination Stop and travel date.

2. The system validates that both Stops are served, the origin precedes the destination and the date is not in the past.

3. The system retrieves Scheduled Trips matching the criteria whose booking cutoff has not passed.

4. The system displays route, selected Stops, planned boarding/alighting times, vehicle type and fare for the selected Segment, ordered by boarding time ascending.

5. Passenger reviews the results and may select a Trip for UC-P05.

4.2.2 Alternative Flows

4.2.2.1 A1 — Invalid search criteria

A1.1 At Step 2, criteria are missing, the date is in the past, or the Stop selection is invalid.

A1.2 The system identifies the correction and returns to Step 1; no Booking is created.

4.2.2.2 A2 — No matching sellable Trip

A2.1 At Step 3, no matching Trip is Scheduled and still open for booking.

A2.2 The system displays no results and permits revised criteria at Step 1.

4.2.2.3 A3 — Trip records cannot be retrieved

A3.1 At Step 3, records cannot be loaded.

A3.2 The system reports a retrieval error rather than inventing results; records remain unchanged.

4.3 Special Requirements

4.3.1 Business Rules

[BR-P04-01] UC-P04 is public and does not require login. Cancelled, Departed and Completed Trips are excluded from new-booking results.

[BR-P04-02] New Bookings close 30 minutes before planned departure at the Route's first Stop; the same cutoff applies to every Segment of that Trip.

[BR-P04-03] The Segment fare is the sum of configured fares for consecutive legs between boarding and alighting Stops. All seats on the same Trip use the same fare for that Segment.

[BR-P04-04] Displayed times and fares come from operator-managed Trip records.

[BR-P04-05] Travel date means the calendar date of the selected boarding Stop's planned time in Asia/Ho_Chi_Minh, including overnight intermediate boarding. Convert timestamps to that zone before comparing dates; reject past selected boarding dates. Keep the Trip-wide sales cutoff at 30 minutes before the first Stop's planned departure. Staff departure lists and schedule-generation dates still use the first Stop date; label these dates distinctly.

4.3.2 Approved Decisions and Scope

Public search, no past-date search, Scheduled-only sale and one Trip-wide 30-minute cutoff are approved. Dates/times use Asia/Ho_Chi_Minh. DEC-01 option A was approved by the project lead on 4 November 2026: Passenger search and AI search use the selected boarding Stop's calendar date, including overnight boarding (BR-P04-05). The common first-Stop cutoff is unchanged. Vision NFR02 defines the performance benchmark.

Specification owner: Nguyễn Khánh Hoàng | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026; DEC-01 date-anchor amendment approved: 4 November 2026.

4.4 Preconditions

4.4.1 Required System and Actor State

Public trip search is available. The operator may have zero or more Trips.

4.5 Postconditions

4.5.1 Success

Matching sellable Trips or an informative empty result are displayed; no allocation or Booking is created.

4.5.2 Failure and Minimal Guarantees

Trip, Seat, Booking, Payment and Ticket data remain unchanged.

4.6 Extension Points

4.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

5. UC-P05 View Trip Details

5.1 View Trip Details

5.1.1 Brief Description

See the full information of a selected trip (time, stops, vehicle, fare) to decide whether to book.

Related Feature(s): F04

Primary Actor: Passenger

Supporting Actor(s): None

Trigger: Passenger selects a Trip from ordinary search results, AI Trip Recommendations (UC-P11), or another valid Trip reference.

5.2 Flow of Events

5.2.1 Basic Flow

1. Passenger opens the selected Trip.

2. The system retrieves the current Trip, vehicle, ordered Stops, planned Stop times and consecutive-leg fares.

3. The system determines whether the Trip remains Scheduled, before cutoff, and complete enough to book.

4. The system displays available details and the fare for the selected boarding/alighting Segment; optional missing descriptions are marked as unavailable.

5. Passenger reviews the details and may continue to UC-P07 when the Trip is open for booking.

5.2.2 Alternative Flows

5.2.2.1 A1 — Trip not found

A1.1 At Step 2, the reference no longer resolves to a Trip.

A1.2 The system explains that it is unavailable and returns Passenger to search; no data changes.

5.2.2.2 A2 — Trip no longer sellable

A2.1 At Step 3, the Trip is not Scheduled or the 30-minute booking cutoff has passed.

A2.2 The system shows the current status and disables new booking actions; Passenger may search for another Trip.

5.2.2.3 A3 — Essential information incomplete

A3.1 At Step 3, required Stop/time, vehicle/seat-layout or fare information is missing or invalid.

A3.2 The system identifies that booking is unavailable and does not permit seat reservation. Available descriptive information may still be viewed.

5.2.2.4 A4 — Trip details cannot be retrieved

A4.1 At Step 2, current Trip, vehicle, Stop-time or fare records cannot be retrieved.

A4.2 The system reports a retrieval error, shows no fabricated details and does not offer booking with unverified essential data. Passenger may retry Step 1 or return to search.

5.3 Special Requirements

5.3.1 Business Rules

[BR-P05-01] UC-P05 is public. Essential booking data consists of a valid ordered Stop sequence, planned Stop times, assigned vehicle/seat layout and configured leg fares.

[BR-P05-02] Missing optional descriptions do not prevent viewing; missing essential data prevents booking.

[BR-P05-03] One Segment fare applies to every seat on the same Trip and is calculated by summing its consecutive-leg fares.

5.3.2 Approved Decisions and Scope

Partial display is retained for optional information only. Multiple seat-price classes and booking after departure are outside this version. Trip-detail response is evaluated under Vision NFR02.

Specification owner: Nguyễn Khánh Hoàng | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

5.4 Preconditions

5.4.1 Required System and Actor State

Passenger has selected a Trip reference. Viewing trip information is public and does not require login.

5.5 Postconditions

5.5.1 Success

Current Trip details and booking availability are shown without changing journey records.

5.5.2 Failure and Minimal Guarantees

Unavailable or incomplete Trips cannot be booked through the details page.

5.6 Extension Points

5.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

6. UC-P06 View Electronic Ticket and Booking History

6.1 View Electronic Ticket and Booking History

6.1.1 Brief Description

View the Passenger’s electronic tickets, booking codes, statuses and past bookings.

Related Feature(s): F08

Primary Actor: Passenger

Supporting Actor(s): None

Trigger: The Passenger opens their bookings or tickets.

6.2 Flow of Events

6.2.1 Basic Flow

1. Passenger opens booking history.

2. The system verifies the session and retrieves only Bookings owned by Passenger.

3. The system lists Booking reference, Trip, Segment, seat count and separate Booking, Payment and Refund statuses.

4. Passenger selects one of their Bookings.

5. The system displays Booking details and the individual Tickets, one per successfully confirmed seat.

6. Passenger views each Ticket code, passenger name, Trip/date, seat, boarding/alighting Stops and times, fare and current Ticket status.

7. After owned Booking details are displayed, Passenger may choose Cancel Booking. At EP-P06-CANCEL, UC-P10 extends this use case if Passenger selects cancellation for that Booking; otherwise viewing completes without any change. P10 evaluates its own eligibility and confirmation rules.

6.2.2 Alternative Flows

6.2.2.1 A1 — Authentication required

A1.1 At Step 2, the session is missing or expired.

A1.2 The system shows no protected data and offers UC-P02. After login, the Passenger may resume at Step 1.

6.2.2.2 A2 — No owned Bookings

A2.1 At Step 2, Passenger has no Bookings.

A2.2 The system shows an empty history; no records change.

6.2.2.3 A3 — Booking not owned or not found

A3.1 At Step 5, the requested reference is not an accessible owned Booking.

A3.2 The system denies access without disclosing whether another passenger's Booking exists.

6.2.2.4 A4 — No valid Ticket available

A4.1 At Step 5, the Booking is Pending, Expired or Cancelled, or a Ticket has been cancelled.

A4.2 The system displays the actual records and statuses rather than presenting a usable Ticket. Checked In Tickets are shown as already used.

6.2.2.5 A5 — History or Ticket retrieval fails

A5.1 At Step 2 or 5, the owned Booking/Ticket records cannot be retrieved.

A5.2 The system reports temporary unavailability rather than an empty history or a valid Ticket. Passenger may retry Step 1; no record or status is changed.

6.3 Special Requirements

6.3.1 Business Rules

[BR-P06-01] A Passenger can view only their own Bookings and Tickets.

[BR-P06-02] One confirmed Booking contains one uniquely coded Ticket for each paid seat; each Ticket records that seat's passenger name and Segment.

[BR-P06-03] Ticket fields are Ticket code, Booking reference, passenger name, Trip/date, seat, boarding/alighting Stops and times, Segment fare and Ticket status. QR codes are not required. Show the selected boarding Stop's local date/time explicitly. If the Trip's first-Stop departure date is also displayed, label it separately; the two can differ on overnight journeys.

[BR-P06-04] Booking statuses: Pending, Confirmed, Expired, Cancelled. Payment attempt statuses: Pending, Succeeded, Failed, Cancelled. Reconciliation statuses: Not Required, Review Required, Resolved. Ticket statuses: Valid, Checked In, Cancelled. Refund statuses: Not Required, Pending, Completed, Failed.

[BR-P06-05] Booking, Payment, Ticket and Refund statuses remain separate. Payment success alone does

not guarantee a valid Ticket.

6.3.2 Approved Decisions and Scope

Authentication is mandatory for history and Tickets. Multiple Tickets per Booking, individual Ticket codes and explicit lifecycle statuses are approved.

Specification owner: Nguyễn Khánh Hoàng | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

6.4 Preconditions

6.4.1 Required System and Actor State

Passenger has an active authenticated Passenger session. Booking history may be empty.

6.5 Postconditions

6.5.1 Success

Passenger sees only their own Booking history and the current status of each individual Ticket.

6.5.2 Failure and Minimal Guarantees

No Booking, Ticket, allocation, Payment or Refund is changed or disclosed to another Passenger.

6.6 Extension Points

6.6.1 EP-P06-CANCEL — Booking details displayed

EP-P06-CANCEL occurs after owned Booking details are displayed (Basic Flow Step 7). UC-P10 extends P06 when Passenger chooses Cancel for that Booking. Viewing is a complete goal without this extension; eligibility, confirmation and all state changes belong to P10.

7. UC-P07 View Seat Availability

7.1 View Seat Availability

7.1.1 Brief Description

Allow a Passenger to view which physical seats are available on a selected Trip for the chosen boarding and alighting Stops.

Related Feature(s): F05

Primary Actor: Passenger

Supporting Actor(s): None

Trigger: Passenger requests the seat map and availability for a Trip and travel Segment.

7.2 Flow of Events

7.2.1 Basic Flow

1. Passenger requests the seat map for the selected Trip and Segment.

2. The system validates the Trip's Scheduled status, the 30-minute booking cutoff, the assigned seat layout and boarding-before-alighting Stop order.

3. The system retrieves seats and effective allocations on the selected Segment.

4. The system treats Confirmed Bookings and unexpired Seat Holds on overlapping Segments as unavailable; expired or cancelled allocations do not block seats.

5. The system displays current Segment-specific availability and updates it when a displayed hold expires.

6. Passenger reviews seats and may proceed to UC-P08, which requires login and a new availability check.

7.2.2 Alternative Flows

7.2.2.1 A1 — Trip or Segment invalid

A1.1 At Step 2, the Trip, layout or Stops are invalid, the Trip is no longer Scheduled or sales have closed.

A1.2 The system explains the selection problem and offers a different Trip or Segment without changing allocations.

7.2.2.2 A2 — No available seats

A2.1 At Step 5, every seat has an overlapping effective allocation.

A2.2 The system displays no available seats and offers another Trip or Segment.

7.2.2.3 A3 — Availability unavailable

A3.1 At Step 3, allocation or seat-layout records cannot be retrieved.

A3.2 The system reports that availability is unknown rather than displaying seats as free.

7.2.2.4 A4 — Hold expires while viewing

A4.1 At Step 4 or 5, an effective Seat Hold reaches its server-defined expiry.

A4.2 The system excludes the expired hold, updates the displayed map and marks the associated pending Booking Expired. Viewing does not acquire the released seat.

7.3 Special Requirements

7.3.1 Business Rules

[BR-P07-01] A Seat Hold lasts 10 minutes from successful creation, using the system's authoritative time; refreshes do not restart it.

[BR-P07-02] Only Confirmed Bookings and unexpired holds block overlapping Segments. A→B and B→C are compatible; A→C and B→C conflict for the same Trip seat.

[BR-P07-03] The map updates after hold expiry, but the display is informational: availability is always checked again when creating a hold.

[BR-P07-04] Public viewing does not reserve any seat. Booking requires an authenticated Passenger.

7.3.2 Approved Decisions and Scope

Ten-minute holds from successful creation, an expiry countdown/update and public viewing are approved. Server expiry remains authoritative if a countdown or page is stale.

Specification owner: Trương Đình Nhật Huy | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

7.4 Preconditions

7.4.1 Required System and Actor State

Passenger has selected a Trip reference and boarding/alighting Stops. Viewing availability is public.

7.5 Postconditions

7.5.1 Success

Passenger sees Segment-specific availability. Any expired hold no longer blocks a seat; no new allocation is created.

7.5.2 Failure and Minimal Guarantees

Unknown availability is not presented as free. Failed viewing does not create or alter an effective allocation.

7.6 Extension Points

7.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

8. UC-P08 Create Temporary Booking

8.1 Create Temporary Booking

8.1.1 Brief Description

Allow a Passenger to select one or more seats for a valid Trip Segment and obtain a temporary reservation pending payment.

Related Feature(s): F06

Primary Actor: Passenger

Supporting Actor(s): None

Trigger: Passenger submits selected seats and journey details for booking.

8.2 Flow of Events

8.2.1 Basic Flow

1. Passenger selects a Trip, boarding and alighting Stops, and one to six seats.

2. The system verifies the Passenger session, Scheduled Trip status, booking cutoff, ordered Segment and seat identifiers.

3. Passenger provides the name for each seat's traveller and checks the account's full name, email and telephone contact details.

4. The system validates the required information and rechecks every requested seat against overlapping Confirmed Bookings and unexpired holds.

5. The system presents the seat/passenger details, Segment times, per-seat fare and total; Passenger confirms.

6. The system establishes one Pending Booking and a 10-minute Seat Hold for all selected seats on the same Trip and Segment, without allowing a conflicting allocation.

7. The system displays the Booking reference, expiry countdown and next payment action.

8.2.2 Alternative Flows

8.2.2.1 A1 — Login required

A1.1 At Step 2, the Passenger session is missing or expired.

A1.2 The system offers UC-P02 and creates no Booking or hold. After login, selections must be validated again.

8.2.2.2 A2 — Invalid selection or information

A2.1 At Step 2 or 4, the selection violates the one-to-six distinct-seat limit (including a repeated physical seat identifier), Segment order, cutoff, Trip status or required passenger/contact rules.

A2.2 The system identifies the correction and returns to Step 1 or 3; no allocation is created. Repeated seats are rejected, not silently removed to create a smaller Booking.

8.2.2.3 A3 — Seat conflict or concurrent request

A3.1 At Step 4 or 6, a requested seat is no longer available on an overlapping Segment.

A3.2 The system does not create a conflicting or partially successful hold; it refreshes availability and asks Passenger to choose again.

8.2.2.4 A4 — Booking creation fails

A4.1 At Step 6, the Booking and holds cannot be completed together.

A4.2 The system reports failure and leaves no effective partial allocation.

8.2.2.5 A5 — Hold expires

A5.1 After Step 6, 10 minutes elapse before an eligible payment is confirmed.

A5.2 The Booking becomes Expired and its holds stop blocking seats. The system informs Passenger; a new Booking requires another availability check.

8.3 Special Requirements

8.3.1 Business Rules

[BR-P08-01] Creating a Booking requires an authenticated Passenger. One Booking covers one Trip, one Segment and one to six seats, with a traveller name for every seat.

[BR-P08-02] New Bookings are rejected from 30 minutes before planned departure at the first Route Stop, regardless of the selected boarding Stop.

[BR-P08-03] The same per-seat Segment fare applies to all selected seats. Contact details are checked before confirmation and the confirmed fare is preserved for payment.

[BR-P08-04] The 10-minute hold starts only when creation succeeds; reloads, repeated checkout and payment retries never extend it.

[BR-P08-05] Concurrent requests must not create overlapping effective allocations. Expired holds do not authorize confirmation or Ticket issuance.

8.3.2 Approved Decisions and Scope

One to six seats, per-seat traveller names, account contact details, fixed 10-minute expiry and the first-Stop booking cutoff are approved.

Specification owner: Trương Đình Nhật Huy | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

8.4 Preconditions

8.4.1 Required System and Actor State

Passenger is authenticated. The operator has a Trip with a valid vehicle layout, ordered Stops, planned Stop times and leg fares.

8.5 Postconditions

8.5.1 Success

A Pending Booking owned by Passenger holds all selected seats and the Segment until its fixed expiry. No Ticket is issued yet.

8.5.2 Failure and Minimal Guarantees

No overlapping or partial effective allocation remains after failure. Expired holds do not block later Bookings.

8.6 Extension Points

8.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

9. UC-P09 Make Test Payment

9.1 Make Test Payment

9.1.1 Brief Description

Allow a Passenger to complete a test payment for a pending Booking and receive a correctly verified Booking/Payment outcome.

Related Feature(s): F07

Primary Actor: Passenger

Supporting Actor(s): VNPay Sandbox (planned for PA5)

Trigger: Passenger selects the payment action for a pending Booking.

9.2 Flow of Events

9.2.1 Basic Flow

1. Passenger requests payment for the Booking.

2. The system verifies ownership, Pending status, unexpired hold, exact saved amount, fewer than three initiated attempts and no other Pending payment attempt.

3. The system creates a uniquely referenced payment attempt and opens the clearly identified test checkout for that amount.

4. Passenger completes the checkout interaction.

5. The system obtains the authoritative payment result and verifies its identity, amount, status and provider authenticity on the server when VNPay Sandbox is integrated.

6. The system records the attempt result and rechecks Booking status, hold validity and seat-allocation eligibility.

7. For the first verified successful eligible payment, the system confirms the Booking, replaces holds with Confirmed allocations and issues exactly one uniquely coded Valid Ticket for each paid seat.

8. The system displays the separate Booking, Payment, reconciliation and Ticket outcomes.

9.2.2 Alternative Flows

9.2.2.1 A1 — Not owned, not payable or expired

A1.1 At Step 2, authentication/ownership fails, the Booking is not Pending or its hold has expired.

A1.2 The system denies checkout without exposing another Passenger's records. Expired holds cannot produce Tickets.

9.2.2.2 A2 — Retry not permitted

A2.1 At Step 2, another attempt is Pending or three attempts have already been initiated.

A2.2 The system explains that a second simultaneous attempt or further retry is unavailable; it does not restart the hold.

9.2.2.3 A3 — Failed or cancelled checkout

A3.1 At Step 4 or 5, the authoritative outcome is Failed or Cancelled.

A3.2 The system records that attempt outcome. A further attempt is allowed only within the original hold and three-attempt limit; no Ticket is issued.

9.2.2.4 A4 — Unverified, inconsistent or unavailable outcome

A4.1 At Step 5, authenticity, identity or amount fails validation, or no authoritative outcome is available.

A4.2 The system does not record a verified success or issue Tickets; the attempt remains Pending until an authoritative terminal result and is flagged Review Required when unresolved or inconsistent. A client redirect alone is insufficient proof.

9.2.2.5 A5 — Duplicate notification

A5.1 At Step 5 or 7, a previously processed notification arrives again.

A5.2 The system preserves the recorded result and does not repeat confirmation, allocation changes or Ticket issuance.

9.2.2.6 A6 — Verified success after expiry, cancellation or allocation conflict

A6.1 At Step 6, money is verified as successfully paid but the Booking/hold is no longer eligible.

A6.2 The system records Payment Succeeded and reconciliation Review Required without confirming the Booking or issuing Tickets. The operator's external back-office process handles full repayment of the unmatched amount.

9.2.2.7 A7 — More than one successful transaction

A7.1 At Step 6, a separate successful transaction is received after an accepted successful payment already confirmed the Booking.

A7.2 The system keeps the existing Tickets and confirmation, flags the extra transaction Review Required and requests full repayment through the external back-office process; no additional Ticket is issued.

9.3 Special Requirements

9.3.1 Business Rules

[BR-P09-01] Every attempt has its own reference and status: Pending, Succeeded, Failed or Cancelled. At most one attempt is Pending at a time, and at most three attempts may be initiated for one Booking.

[BR-P09-02] An unknown result is not a failed result and does not authorize a parallel retry. Payment retry does not extend the original 10-minute hold or change the amount.

[BR-P09-03] Reconciliation has separate statuses Not Required, Review Required and Resolved. Review flags do not replace the Payment outcome.

[BR-P09-04] Only the first verified successful payment while the Booking is Pending and its hold/allocations remain eligible may confirm it. Exactly one Ticket is issued per paid seat, even when callbacks repeat.

[BR-P09-05] Verified money paid too late, after cancellation, or in an extra transaction requires external reconciliation and full repayment of the unmatched amount; it never recreates a lost seat allocation.

[BR-P09-06] PA4 may use a clearly labelled mock to demonstrate simulated Booking/Ticket outcomes with test data. Mock outcomes are not provider-verified payments and involve no real money. PA5 integrates VNPay Sandbox with server-side result verification.

9.3.2 Approved Decisions and Scope

Three attempts, one Pending attempt at a time, no hold extension and external reconciliation/repayment are approved. UC-A04 displays reconciliation information but provides no manual approval or

refund-execution controls.

Specification owner: Trương Đình Nhật Huy | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

9.4 Preconditions

9.4.1 Required System and Actor State

Passenger is authenticated and owns a Pending Booking with a saved amount and selected seats/Segment. A clearly labelled test-payment mechanism is available for the project phase.

9.5 Postconditions

9.5.1 Success

One eligible verified successful payment confirms the Booking and creates one Valid Ticket per paid seat exactly once. Payment and reconciliation outcomes are separately recorded.

9.5.2 Failure and Minimal Guarantees

Failed, unknown, unverified, late or extra outcomes never create unauthorized Tickets or conflicting allocations. An accepted earlier result is preserved.

9.6 Extension Points

9.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

10. UC-P10 Cancel Booking

10.1 Cancel Booking

10.1.1 Brief Description

Allow a Passenger to cancel an eligible Booking and release exactly the seat allocations belonging to the cancelled journey Segment.

Related Feature(s): F09

Primary Actor: Passenger

Supporting Actor(s): None

Trigger: Passenger requests cancellation directly or chooses Cancel at P06 extension point EP-P06-CANCEL after viewing an owned Booking.

10.2 Flow of Events

10.2.1 Basic Flow

1. Passenger requests cancellation of a Booking.

2. The system verifies the session, ownership and current Booking status.

3. The system checks the cancellation policy: Pending Bookings may be cancelled; Confirmed Bookings require at least 12 hours before planned boarding at the selected Stop and no Checked In Ticket.

4. The system shows that the entire Booking will be cancelled, affected seats/Tickets and the refund amount: zero for an unpaid Pending Booking or 90 percent of the paid ticket fare for an eligible Confirmed Booking.

5. Passenger explicitly confirms cancellation.

6. The system rechecks eligibility and cancels the complete Booking, releases only its held/confirmed Segments and marks all its Tickets Cancelled.

7. For a paid Confirmed Booking, the system records Refund Pending for 90 percent of the paid ticket fare; payment outcomes remain recorded separately. The operator's back-office process executes the refund outside this application.

8. The system displays cancellation and refund statuses separately.

10.2.2 Alternative Flows

10.2.2.1 A1 — Unauthorized or missing Booking

A1.1 At Step 2, the Booking is missing or not owned by Passenger.

A1.2 The system rejects the action without disclosing another Passenger's records; no data changes.

10.2.2.2 A2 — Cancellation ineligible

A2.1 At Step 3 or 6, a Confirmed Booking is less than 12 hours before its planned boarding time, any Ticket is Checked In, or the Booking is Expired.

A2.2 The system explains that self-service cancellation is unavailable and does not change Booking, Ticket or allocations.

10.2.2.3 A3 — Already cancelled

A3.1 At Step 2, the Booking is already Cancelled.

A3.2 The system displays its existing cancellation/refund status without releasing seats again or creating another refund request.

10.2.2.4 A4 — Passenger declines

A4.1 At Step 5, Passenger does not confirm.

A4.2 The system returns to Booking details with all states unchanged.

10.2.2.5 A5 — Cancellation cannot be saved

A5.1 At Step 6 or 7, the cancellation operation cannot be completed consistently.

A5.2 The system reports failure rather than a false success; it does not leave released seats paired with valid Tickets or create duplicate refund requests.

10.2.2.6 A6 — Pending payment later succeeds

A6.1 After a Pending Booking has been cancelled, a payment outcome arrives.

A6.2 UC-P09 processes it against the cancelled Booking: no Ticket is issued; a verified paid amount is flagged for external full repayment.

10.3 Special Requirements

10.3.1 Business Rules

[BR-P10-01] Passenger cancellation applies to the entire Booking; partial seat cancellation is outside scope.

[BR-P10-02] A Pending Booking may be cancelled at any time before confirmation. A Confirmed Booking may be cancelled only when at least 12 hours remain before planned boarding at the Passenger's selected Stop and no Ticket has been checked in.

[BR-P10-03] Eligible Passenger cancellation refunds 90 percent of the paid ticket fare; the remaining 10 percent is the cancellation charge. Cancellation of an unpaid Booking has Refund Not Required.

[BR-P10-04] Operator cancellation of a Scheduled Trip is handled by UC-S04, invalidates Tickets and requests a 100-percent refund of the paid fare regardless of the Passenger cancellation cutoff.

[BR-P10-05] Refund statuses are Not Required, Pending, Completed and Failed. Refund execution and result recording belong to a controlled operator back-office process outside the 20 interactive use cases; the demo uses test records to display these outcomes.

[BR-P10-06] Cancellation releases only this Booking's Seat/Segment allocations. Cancellation success never implies that repayment has completed.

[BR-P10-07] Compute a whole-Booking passenger refund once from the nonnegative integer paid fare F in VND: refundVND = floor((9 × F + 5) / 10), equivalent to rounding 90 percent to the nearest whole VND with halves rounded up. cancellationChargeVND = F − refundVND. Do not round each seat separately. Operator 100-percent refunds and full unmatched-payment repayment equal the original paid amount.

10.3.2 Approved Decisions and Scope

The 12-hour limit uses the selected boarding Stop's planned time. Whole-Booking cancellation, the 90-percent passenger refund and external refund execution remain approved; automated refund integration and partial cancellation are excluded. DEC-02 option A was approved on 4 November 2026: round the whole-Booking refund to nearest whole VND, halves up, then derive the cancellation charge as the remainder (BR-P10-07).

Specification owner: Trương Đình Nhật Huy | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026; DEC-02 refund-rounding amendment approved: 4 November 2026.

10.4 Preconditions

10.4.1 Required System and Actor State

Passenger is authenticated. The application can evaluate the approved whole-Booking cancellation policy and access the Passenger's records.

10.5 Postconditions

10.5.1 Success

The entire eligible Booking is Cancelled, its allocations are released and all its Tickets are Cancelled. A paid passenger cancellation creates one Refund Pending request for 90 percent.

10.5.2 Failure and Minimal Guarantees

Unauthorized or ineligible requests leave records unchanged. Neither a failed operation nor a duplicate request produces false cancellation or a duplicate refund.

10.6 Extension Points

10.6.1 None Defined

This use case defines no extension point of its own. It extends UC-P06 at EP-P06-CANCEL when Passenger chooses cancellation from owned Booking details. Direct authorized entry to P10 remains available; P06 does not include cancellation.

11. UC-P11 Get AI Trip Recommendations

11.1 Get AI Trip Recommendations

11.1.1 Brief Description

Allow a Passenger to describe a journey in natural language and receive recommendations made up only of real Trips from the operator's records that match validated search criteria.

Related Feature(s): F03

Primary Actor: Passenger

Supporting Actor(s): External AI Service (Google Gemini Developer API). Passenger remains the primary actor.

Trigger: Passenger submits a trip request to the AI Trip Assistant.

11.2 Flow of Events

11.2.1 Basic Flow

1. Passenger opens the AI Trip Assistant.

2. Passenger describes origin, destination, date, time range, maximum budget or vehicle-type preference in Vietnamese.

3. The system interprets the request as structured trip-search criteria.

4. The system validates the criteria against supported filters and operator-managed Stops, dates and vehicle types.

5. The system retrieves matching Scheduled Trips still before the common booking cutoff, applying the selected Segment's fare and Stop times.

6. The system displays at most five matching Trips ordered by planned boarding time ascending, with applied criteria, Trip/vehicle summaries and Segment fares.

7. Passenger reviews the results and may open UC-P05 or UC-P07 for current seat availability.

11.2.2 Alternative Flows

11.2.2.1 A1 — Request unclear or incomplete

A1.1 At Step 3 or 4, the system cannot reliably identify required Stops/date or a preference is ambiguous.

A1.2 The system shows understood criteria and requests clarification; it does not invent values. The use case resumes at Step 2.

11.2.2.2 A2 — Invalid or unsupported criteria

A2.1 At Step 4, a Stop/date/Segment or preference is invalid or unsupported, or the request is not written in Vietnamese (BR-P11-01).

A2.2 The system identifies the unsupported criterion and lets Passenger revise the request at Step 2.

11.2.2.3 A3 — No matching Trip

A3.1 At Step 5, no actual sellable Trip satisfies the validated filters.

A3.2 The system shows no matching Trips and offers revised criteria or UC-P04; it does not fabricate an alternative.

11.2.2.4 A4 — AI unavailable, timeout or limit reached

A4.1 At Step 3, the service fails, 10 seconds elapse, five requests in one minute are exceeded for the user/session, or the demo's 200-request daily cap is reached.

A4.2 The system reports temporary unavailability or the limit and offers UC-P04. No journey state changes.

11.2.2.5 A5 — Trip records unavailable

A5.1 At Step 5, current records cannot be retrieved.

A5.2 The system reports the error and does not present generated text as real Trip results.

11.2.2.6 A6 — Action or private-data request outside scope

A6.1 At Step 2 or 3, Passenger asks the assistant to book, pay, cancel, or reveal another user's Booking.

A6.2 The system explains the recommendation-only scope and points to the authorized normal function; no

protected data is disclosed or journey action performed.

11.3 Special Requirements

11.3.1 Business Rules

[BR-P11-01] Supported request language is Vietnamese. Filters are origin, destination, date, time range, budget and vehicle type. Unknown criteria require clarification or an explicit unsupported-filter response. Date filters follow BR-P04-05: use the selected boarding Stop's Vietnam-local calendar date rather than the first-Stop departure date.

[BR-P11-02] Only actual operator-managed Scheduled Trips before the 30-minute first-Stop cutoff may be returned. Return at most five, ordered by boarding time ascending.

[BR-P11-03] Recommendations do not display or guarantee individual-seat availability, create holds or reserve seats; UC-P07 provides current availability and UC-P08 checks again.

[BR-P11-04] AI output is validated search criteria, not authoritative Trip data. Actual results are retrieved by the application from operator records.

[BR-P11-05] Under leader-approved QĐ-07 (8 November 2026), Google Gemini Developer API replaces OpenAI as the AI provider. The demo targets the Free Tier with the model configured in the backend; availability and project quota must be checked before the PA4 demo. No automatic paid fallback.

[BR-P11-06] AI limits are five requests per minute per authenticated user or anonymous session and 200 requests per Vietnam-local calendar day for the demo, shared by P11/P12. A request unresolved after 10 seconds receives a fallback.

[BR-P11-07] Conversation history is not persisted in the application's business database. Full names, email, telephone numbers and whole Booking records are not sent to the AI provider.

11.3.2 Approved Decisions and Scope

Vietnamese input, five-result ordering, supported preferences, shared application limits, timeout and application-side conversation non-retention remain approved. QĐ-07 replaces the provider with Google Gemini Developer API Free Tier. Model/key are backend configuration; free-tier provider data handling is assessed in SAD, without a zero-retention claim.

Specification owner: Đinh Đức Hiếu | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

11.4 Preconditions

11.4.1 Required System and Actor State

The public AI Trip Assistant is available and the operator's Trip records may be empty. Login is not required.

11.5 Postconditions

11.5.1 Success

Passenger sees up to five real Trips matching validated criteria; no Booking, Payment, Ticket or allocation is created.

11.5.2 Failure and Minimal Guarantees

No invented Trip/price/availability or protected Booking data is shown. Ordinary search remains available independently of AI.

11.6 Extension Points

11.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do

not imply an «extend» relationship.

12. UC-P12 Ask AI Customer Support

12.1 Ask AI Customer Support

12.1.1 Brief Description

Allow a Passenger to ask questions in natural language and receive answers based only on approved policy information and, for questions about a specific Booking, on Trip and Booking data the Passenger is allowed to see.

Related Feature(s): F10

Primary Actor: Passenger

Supporting Actor(s): External AI Service (Google Gemini Developer API). Passenger remains the primary actor.

Trigger: Passenger submits a question to AI Customer Support.

12.2 Flow of Events

12.2.1 Basic Flow

1. Passenger opens AI Customer Support.

2. Passenger submits a general policy question in Vietnamese.

3. The system classifies it as a supported general policy topic; Trip and Booking requests follow A8 or A9.

4. The system retrieves the current approved FAQ/policy content for that topic.

5. The system prepares an answer grounded only in that approved content.

6. The system displays the answer and identifies the policy topic used.

7. Passenger reviews the answer; no journey, Payment, Ticket or Refund record is changed.

8. Passenger may ask another question at Step 2 or leave the conversation.

12.2.2 Alternative Flows

12.2.2.1 A1 — Question unclear or Booking unspecified

A1.1 At Step 3, the question is ambiguous or not written in Vietnamese, or the Trip/Booking reference needed by A8/A9 is unspecified.

A1.2 The system shows what it understood and asks Passenger to clarify at Step 2 or the indicated A8/A9 reference step; it does not guess a record.

12.2.2.2 A2 — No approved answer

A2.1 At Step 4 or the retrieval step of A8/A9, no approved policy or supported factual answer exists.

A2.2 The system says that it cannot answer and displays the operator contact channel from application configuration; it does not invent a policy or contact value.

12.2.2.3 A3 — AI unavailable, timeout or limit reached

A3.1 At Step 3 or 5, or while preparing an A8 answer, the AI service fails, the 10-second response limit or the shared request cap is reached.

A3.2 The system reports unavailability and offers the relevant normal function, such as UC-P04 or UC-P06; records remain unchanged.

12.2.2.4 A4 — Booking authentication missing

A4.1 On the Booking branch A9 entered from Step 3, the Passenger session is missing or expires before protected information is displayed.

A4.2 The system offers UC-P02 and shows no Booking information. After login it restarts A9 with fresh ownership checks; general policy questions remain public.

12.2.2.5 A5 — Booking inaccessible

A5.1 On the Booking branch A9 entered from Step 3, the reference is missing, inaccessible or not owned by Passenger.

A5.2 The system gives no Booking details and does not confirm another user's record existence; Passenger may ask another question.

12.2.2.6 A6 — Action requested

A6.1 At Step 3, Passenger asks support to reserve/change a seat, cancel, confirm payment, issue a Ticket or execute a refund.

A6.2 The system explains its information-only scope and points to an appropriate authorized use case; no record is modified.

12.2.2.7 A7 — Information cannot be retrieved

A7.1 At Step 4 or a retrieval step of A8/A9, the approved FAQ or current record cannot be retrieved.

A7.2 The system reports the failure rather than displaying guessed or unverified current states.

12.2.2.8 A8 — Question about an identified Trip

A8.1 At Step 3, Passenger asks a factual question about one Trip rather than a general policy. The system requests a Trip reference if it is not already identified.

A8.2 The system validates the reference and retrieves authorized public Trip facts; an unclear reference follows A1, an unsupported answer A2 and a retrieval failure A7.

A8.3 The system presents only those current facts with the Trip reference, then resumes at Step 8. Trip recommendations remain UC-P11.

12.2.2.9 A9 — Question about an owned Booking

A9.1 At Step 3, Passenger asks about a specific Booking. The system requests its reference and verifies an active Passenger session. Missing authentication follows A4.

A9.2 The system verifies ownership before retrieving or revealing protected information. An inaccessible record follows A5.

A9.3 The application retrieves and displays the authorized Booking facts, with Booking, individual Ticket, Payment, reconciliation and Refund states shown separately. The whole Booking and identifying contact data are not sent to the AI provider; retrieval failure follows A7.

A9.4 The system identifies the Booking reference used and resumes at Step 8 without modifying records.

12.3 Special Requirements

12.3.1 Business Rules

[BR-P12-01] UC-P12 serves Passenger. General policy and factual Trip questions are public; personal Booking questions require login and ownership. Staff/Admin-specific support is excluded.

[BR-P12-02] FAQ topics are booking, Seat Holds, payment, cancellation/refunds and check-in. The team prepares content, the project Leader approves it for the demo, and the operator's representative is the business policy approver.

[BR-P12-03] FAQ content is prepared outside the application; FAQ-management screens and live handoff to Staff are outside scope. Unknown answers show the operator-configured official contact channel.

[BR-P12-04] P12 answers factual questions about an identified Trip; P11 searches for and recommends Trips.

[BR-P12-05] Support is read-only. Only approved FAQ and authorized current records may support an answer. Private Booking results are authorization-checked and rendered by the application; personal names, email, telephone numbers and whole Booking records are not forwarded to the AI provider.

[BR-P12-06] Use Google Gemini Developer API under QĐ-07, with the application limits shared with P11: five requests per minute per user/session, 200 per Vietnam-local demo day and a ten-second fallback. Provider limits are additional and project/model-specific; quota exhaustion returns fallback. No persistent application conversation history; no private Booking facts or personal data sent to the free service.

12.3.2 Approved Decisions and Scope

Passenger-only support, public FAQ, owned-Booking authorization, FAQ approval ownership, configured contact fallback, no live handoff and the P11/P12 responsibility split are approved.

Specification owner: Đinh Đức Hiếu | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

12.4 Preconditions

12.4.1 Required System and Actor State

Public AI policy support is available. Operator-approved FAQ content is maintained; personal Booking questions additionally require an authenticated Passenger.

12.5 Postconditions

12.5.1 Success

Passenger receives an answer grounded in approved content or authorized facts; no Booking, allocation, Ticket, Payment or Refund state changes.

12.5.2 Failure and Minimal Guarantees

No unauthorized personal data, invented policy, fabricated current state or unapproved operational action is exposed or performed.

12.6 Extension Points

12.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

13. UC-S01 View Upcoming Departures

13.1 View Upcoming Departures

13.1.1 Brief Description

Allow Staff to view upcoming departures to support passenger and booking operations.

Related Feature(s): F11

Primary Actor: Staff

Supporting Actor(s): None

Trigger: Staff requests to view upcoming departures.

13.2 Flow of Events

13.2.1 Basic Flow

1. Staff opens upcoming departures.

2. The system verifies Staff access and displays today's Trips ordered by planned first-Stop departure time ascending.

3. The system shows Trip code, Route, departure time, assigned vehicle, Trip status, confirmed-issued Ticket count and Checked In Ticket count.

4. Staff may select Route, date/date range or Trip-status filters, with a selected range no longer than seven days.

5. The system validates the filters, displays matching Trips in departure order and lets Staff select a Trip for the related operational function.

13.2.2 Alternative Flows

13.2.2.1 A1 — Unauthorized or expired session

A1.1 At Step 2, Staff access is invalid.

A1.2 The system denies access, clears protected display data and offers login where applicable.

13.2.2.2 A2 — Invalid filter range

A2.1 At Step 5, dates are invalid or the chosen range exceeds seven days.

A2.2 The system requests corrected filters and resumes at Step 4.

13.2.2.3 A3 — No departures or retrieval failure

A3.1 At Step 2 or 5, no Trip matches or records cannot be loaded.

A3.2 The system distinguishes an empty result from a retrieval error and does not invent a departure.

13.3 Special Requirements

13.3.1 Business Rules

[BR-S01-01] Default date is today in Vietnam local time. Staff may choose a range of up to seven days and filter by Route, date and Trip status. This operational departure-date filter uses the Route's first-Stop planned departure date; it is distinct from the Passenger boarding-date search in BR-P04-05.

[BR-S01-02] Display Trip code, Route, first-Stop departure time, assigned vehicle, Trip status, confirmed-issued Ticket count (Valid plus Checked In) and Checked In count. Cancelled Tickets are excluded from those counts.

[BR-S01-03] Results are ordered by departure time ascending. This use case does not update Trip or Booking state.

13.3.2 Approved Decisions and Scope

Today's default view, seven-day range, listed columns/filters and ascending departure order are approved.

Specification owner: Đỗ Minh Phương | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

13.4 Preconditions

13.4.1 Required System and Actor State

Staff has an active Staff session. Trip records may be empty.

13.5 Postconditions

13.5.1 Success

Staff sees the authorized operational Trip list and selected filters without modifying journey records.

13.5.2 Failure and Minimal Guarantees

Unauthorized access reveals no protected data. Empty/error states cause no journey changes.

13.6 Extension Points

13.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

14. UC-S02 Look Up Booking

14.1 Look Up Booking

14.1.1 Brief Description

Allow Staff to retrieve and view booking or ticket information using a booking reference, ticket code, or passenger information for authorized operational purposes.

Related Feature(s): F11

Primary Actor: Staff

Supporting Actor(s): None

Trigger: Staff requests to look up a booking or ticket.

14.2 Flow of Events

14.2.1 Basic Flow

1. Staff opens Booking/Ticket lookup.

2. The system verifies Staff permission and requests a Booking code, Ticket code or contact telephone number.

3. Staff submits one of the supported lookup values.

4. The system validates it and retrieves permitted matching records.

5. The system lists matching Booking/Ticket references, passenger/contact name, Trip/date, Segment and current statuses; multiple matches remain separate records.

6. Staff selects the relevant result.

7. The system displays the permitted passenger name/contact telephone, Trip, boarding/alighting Segment, seats and Booking/Ticket/Payment statuses for operational review.

14.2.2 Alternative Flows

14.2.2.1 A1 — Unauthorized access

A1.1 At Step 2, the session/role is invalid.

A1.2 The system denies lookup without disclosing records.

14.2.2.2 A2 — Invalid input

A2.1 At Step 4, the lookup value is incomplete or invalid.

A2.2 The system requests correction at Step 3.

14.2.2.3 A3 — No match

A3.1 At Step 4, no permitted record matches.

A3.2 The system displays a no-results message and lets Staff try another reference.

14.2.2.4 A4 — Record changed or retrieval failed

A4.1 At Step 4 or 7, a record cannot be retrieved or changed after selection.

A4.2 The system refreshes the authorized result if available or reports the error; it does not display fabricated current data.

14.3 Special Requirements

14.3.1 Business Rules

[BR-S02-01] Lookup keys are Booking code, Ticket code or contact telephone number. Multiple matches are listed for explicit Staff selection; the system never silently chooses one.

[BR-S02-02] Staff may see passenger names, necessary contact telephone, Trip/date, Segment, seats and operational Booking/Ticket/Payment statuses. Account passwords, authentication data and payment-provider technical data are excluded.

[BR-S02-03] Lookup data is cleared when Staff leaves the screen, logs out or reaches the 15-minute Staff inactivity timeout.

[BR-S02-04] Lookup is read-only and does not itself check a passenger in.

14.3.2 Approved Decisions and Scope

The three lookup keys, explicit selection among matches, limited visible fields and screen/session clearing behavior are approved.

Specification owner: Đỗ Minh Phương | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

14.4 Preconditions

14.4.1 Required System and Actor State

Staff has an active Staff session. Lookup is restricted to the single operator's records.

14.5 Postconditions

14.5.1 Success

Staff reviews the intended authorized Booking/Ticket record and its current operational details.

14.5.2 Failure and Minimal Guarantees

No lookup operation changes Booking, Ticket, Payment or allocation states or reveals excluded account/payment data.

14.6 Extension Points

14.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

15. UC-S03 Check In Passenger

15.1 Check In Passenger

15.1.1 Brief Description

Allow authorized staff to validate a passenger's booking or ticket for a selected trip and record check-in when the passenger is eligible.

Related Feature(s): F12

Primary Actor: Staff

Supporting Actor(s): None

Trigger: Staff enters a booking reference or ticket code to check in a passenger for a selected trip.

15.2 Flow of Events

15.2.1 Basic Flow

1. Staff selects the Trip for passenger boarding.

2. Staff enters an individual Ticket code.

3. The system invokes included UC-C01 Validate Ticket for Check-in with the selected Trip and Ticket reference; C01 verifies Staff permission and retrieves current Ticket/Booking/Trip facts.

4. UC-C01 returns Eligible only for a Valid Ticket, Confirmed Booking, matching Trip and permitted boarding Stop. Scheduled is eligible; Departed is eligible only for intermediate boarding. Cancelled/Completed Trips are rejected. A failure returns the corresponding reason to A1–A4/A6.

5. The system displays the traveller, seat, boarding/alighting Stops, planned boarding time and validation result. Staff reviews that the traveller is boarding at the Ticket's recorded Stop.

6. Staff explicitly confirms Check In for this Ticket.

7. The system invokes UC-C01 again immediately before saving, using the same Ticket/Trip/boarding-Stop context and current session. Only an Eligible result permits recording Checked In with Ticket/Booking/Trip references, Staff identity and time; a changed/failed result follows A1–A4/A6.

8. The system confirms check-in for this Ticket only; other Tickets in the same Booking keep their own statuses.

15.2.2 Alternative Flows

15.2.2.1 A1 — Unauthorized access

A1.1 At Step 3 or 7, the session is invalid or the actor is not authorized Staff.

A1.2 The system denies check-in without changing Ticket state.

15.2.2.2 A2 — Ticket not found or wrong Trip

A2.1 At Step 3, 4 or 5, the code is unknown, belongs to another Trip, a Departed Trip Ticket boards at the first Route Stop, or Staff identifies a mismatch with the recorded boarding Stop.

A2.2 The system reports the Trip/boarding-Stop mismatch and records no check-in. Staff may correct the selection at Step 1 or input at Step 2.

15.2.2.3 A3 — Already checked in

A3.1 At Step 4 or 7, the Ticket is already Checked In.

A3.2 The system shows the existing check-in and does not create a second check-in record.

15.2.2.4 A4 — Cancelled, unconfirmed or inactive journey

A4.1 At Step 4 or 7, the Ticket is Cancelled, its Booking is not Confirmed, or the Trip is Cancelled or Completed.

A4.2 The system rejects check-in and displays the recorded reason; a Payment Succeeded record alone is insufficient.

15.2.2.5 A5 — Staff declines confirmation

A5.1 At Step 6, Staff does not confirm.

A5.2 The system returns to lookup without changing the Ticket.

15.2.2.6 A6 — Retrieval or save failure

A6.1 At Step 3 or 7, required records cannot be retrieved or check-in cannot be saved.

A6.2 The system reports failure and does not claim or leave a partially saved successful check-in.

15.2.2.7 A7 — Locate a Ticket using its Booking reference

A7.1 At Step 2, Staff enters a Booking reference instead of an individual Ticket code.

A7.2 The system verifies Staff permission and the Booking association with the selected Trip, then lists its individual Tickets with traveller names, seats and current statuses. Invalid access follows A1; an unknown or wrong-Trip reference follows A2.

A7.3 Staff explicitly selects one Ticket. The system resumes at Step 3 using that Ticket reference and revalidates its current eligibility; selecting the Booking never checks in all its Tickets.

15.3 Special Requirements

15.3.1 Business Rules

[BR-S03-01] Check-in is per Ticket/per seat, not per whole Booking. Each confirmed Booking may contain several independently checked Tickets.

[BR-S03-02] Eligibility requires a Valid Ticket and Confirmed Booking for the matching Trip. Scheduled is allowed; Departed is allowed only for a Ticket boarding at an intermediate Route Stop, never at the first Stop. Staff verifies boarding at the recorded Stop; automated vehicle-location tracking is not required. Cancelled and Completed Trips are rejected.

[BR-S03-03] Staff must explicitly confirm after reviewing validation. Store Ticket, Booking and Trip references, Staff identity and check-in time.

[BR-S03-04] Check-in does not change the recorded successful Payment, create a new Ticket or release a seat. Checked In Tickets cannot be checked in twice.

[BR-S03-05] Check-in correction/reversal is excluded from this version.

15.3.2 Approved Decisions and Scope

Individual Ticket check-in, explicit Staff confirmation, audit fields, intermediate-Stop boarding on Departed Trips and exclusion of reversal are approved.

Specification owner: Đỗ Minh Phương | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

15.4 Preconditions

15.4.1 Required System and Actor State

Staff has an active Staff session. Trip and Ticket records can be checked; selecting a Trip is part of the flow.

15.5 Postconditions

15.5.1 Success

Exactly the selected Ticket becomes Checked In with Staff/time audit information; other Tickets remain unchanged.

15.5.2 Failure and Minimal Guarantees

Invalid or duplicate requests do not create a check-in. Retrieval/save failures do not produce a false successful status.

15.6 Extension Points

15.6.1 None Defined

No extension point is defined. S03 includes UC-C01 Validate Ticket for Check-in before confirmation and invokes it again immediately before saving; C01 checks eligibility and never records check-in itself.

16. UC-S04 Update Trip Status

16.1 Update Trip Status

16.1.1 Brief Description

Allow authorized Staff to update the lifecycle status of a trip.

Related Feature(s): F13

Primary Actor: Staff

Supporting Actor(s): None

Trigger: Staff requests to update the status of a selected trip.

16.2 Flow of Events

16.2.1 Basic Flow

1. Staff opens the selected Trip's status function.

2. The system verifies Staff permission and displays the current Trip status.

3. Staff selects a permitted next status; a cancellation requires a reason.

4. The system validates the transition and its conditions: Scheduled to Departed requires an assigned valid vehicle/Route and planned first-Stop departure time reached; Departed to Completed requires an already departed Trip; Scheduled to Cancelled requires a reason.

5. The system saves the status change and audit data identifying Staff, time, old/new status and any cancellation reason. For cancellation it also cancels affected Bookings/Tickets, releases their holds/allocations and records required refund requests consistently.

6. The system displays the updated status and any cancellation/refund consequences without claiming that refunds have been paid.

16.2.2 Alternative Flows

16.2.2.1 A1 — Unauthorized or missing Trip

A1.1 At Step 2, permission is invalid or the Trip does not exist.

A1.2 The system denies the operation or reports the missing record; no state changes.

16.2.2.2 A2 — Invalid transition or condition

A2.1 At Step 4, the transition is not allowed, departure is too early, vehicle/Route data is invalid or a

cancellation reason is missing.

A2.2 The system explains the condition and requests a valid action; the current status is preserved.

16.2.2.3 A3 — Status changed concurrently

A3.1 At Step 4 or 5, the Trip no longer has the status originally reviewed.

A3.2 The system refreshes it and requires Staff to review a currently permitted transition; no obsolete change is applied.

16.2.2.4 A4 — Status/cancellation update fails

A4.1 At Step 5, the change and required cancellation consequences cannot be saved consistently.

A4.2 The system reports failure rather than a partial success; valid Tickets must not remain usable on a successfully cancelled Trip.

16.3 Special Requirements

16.3.1 Business Rules

[BR-S04-01] Trip statuses are Scheduled, Departed, Completed and Cancelled. Allowed transitions are Scheduled→Departed, Departed→Completed and Scheduled→Cancelled; reverse transitions are prohibited.

[BR-S04-02] Scheduled→Departed requires a valid assigned vehicle/Route and the planned first-Stop departure time reached. Cancelled and Completed are terminal in this version.

[BR-S04-03] Record Staff identity, time and old/new status for every change; cancellation additionally requires a reason.

[BR-S04-04] Operator cancellation invalidates all affected Tickets, including any already Checked In, cancels Pending/Confirmed Bookings, releases holds/allocations and prevents further check-in.

[BR-S04-05] Confirmed Bookings cancelled by the operator request a 100-percent refund of the paid fare without the Passenger 12-hour restriction or cancellation charge. Existing refund requests are reconciled so total repayment never exceeds the amount paid and no duplicate refund is created.

[BR-S04-06] Refund execution remains outside this application. Any payment success arriving after cancellation follows UC-P09 reconciliation rather than issuing Tickets.

16.3.2 Approved Decisions and Scope

The four Trip states, one-way transitions, departure preconditions, audit fields and operator-cancellation consequences are approved. En-route Trip cancellation and status rollback are outside scope.

Specification owner: Đỗ Minh Phương | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

16.4 Preconditions

16.4.1 Required System and Actor State

Staff has an active Staff session and has a Trip reference to review.

16.5 Postconditions

16.5.1 Success

The valid transition and audit are recorded. If Cancelled, allocations are released, Tickets are invalid and appropriate refund requests are recorded once.

16.5.2 Failure and Minimal Guarantees

An invalid/unauthorized/failed update preserves the previous state and never reports a false success or completed refund.

16.6 Extension Points

16.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

17. UC-A01 Manage Vehicles and Seat Layout

17.1 Manage Vehicles and Seat Layout

17.1.1 Brief Description

Allow the Administrator to create and maintain vehicle records using the two predefined seated/sleeper type and seat-layout templates, with valid capacity and preserved used-Trip references.

Related Feature(s): F14

Primary Actor: Administrator

Supporting Actor(s): None

Trigger: The Administrator requests to create a vehicle or maintain an existing vehicle's information.

17.2 Flow of Events

17.2.1 Basic Flow

1. Administrator opens vehicle management.

2. The system verifies permission and lists vehicles and available fixed layout templates.

3. Administrator chooses to create a vehicle.

4. Administrator enters vehicle code, unique registration plate, vehicle type and one of the seated/sleeper seat-layout templates, with capacity matching that layout.

5. The system checks required fields, unique code/plate, unique seat identifiers within the vehicle and seat-count/capacity consistency.

6. Administrator confirms the vehicle details.

7. The system saves the vehicle and its complete layout.

8. The system displays the saved vehicle and confirms success.

17.2.2 Alternative Flows

17.2.2.1 A1 — Maintain existing vehicle

A1.1 At Step 3, Administrator selects an existing vehicle instead.

A1.2 The system loads the selected vehicle. Administrator edits permitted fields; structural layout/capacity changes remain prohibited once assigned to a Trip.

A1.3 The system applies Step 5 validation to the proposed update. Administrator reviews the validated changes and explicitly confirms.

A1.4 The system saves an update to the selected vehicle identity, preserves dependent Trip/seat references, displays the result and ends this alternative; it does not create a second vehicle.

17.2.2.2 A2 — Invalid or duplicate vehicle data

A2.1 At Step 5, required values are missing, code/plate conflicts, seats repeat, or seat count does not match capacity.

A2.2 The system identifies corrections and saves nothing until valid.

17.2.2.3 A3 — Layout assigned to a Trip

A3.1 At Step 5, a proposed structural seat-layout/capacity change affects a vehicle already assigned to a Trip.

A3.2 The system rejects that structural change and preserves all existing seat references; Administrator may create a new vehicle/layout record instead.

17.2.2.4 A4 — Delete or deactivate

A4.1 At Step 3, Administrator requests removal.

A4.2 The system checks dependent Trips and displays the removal consequence. If never referenced, Administrator may confirm deletion; the system removes that record and confirms completion.

A4.3 A referenced vehicle cannot be deleted. Administrator may instead explicitly confirm deactivation for new assignments; the system saves that availability change while preserving historical/current Trips and displays the result. Declining confirmation leaves all records unchanged.

A4.4 The alternative ends after the saved result; unauthorized, missing-record or save failures follow A5.

17.2.2.5 A5 — Unauthorized, missing record or save failure

A5.1 At Step 2, 3 or 7, access is invalid, the selected record is missing or saving fails.

A5.2 The system denies access or reports the error, without applying partial changes or claiming success.

17.3 Special Requirements

17.3.1 Business Rules

[BR-A01-01] Required vehicle data is code, unique registration plate, type, capacity and a complete seat layout. Types/layout choices are the two predefined seated and sleeper templates; a free-form layout editor/type-management screen is excluded.

[BR-A01-02] Seat identifiers are unique within one vehicle and the seat count equals capacity.

[BR-A01-03] Structural layout/capacity changes are prohibited after assignment to any Trip. Deactivation prevents new assignments and must not silently invalidate existing assignments or sold seats.

[BR-A01-04] Delete only a vehicle never referenced by a Trip; preserve used records for history.

17.3.2 Approved Decisions and Scope

Two fixed layout templates, field/uniqueness rules, assignment-based layout protection and deletion only before use are approved.

Specification owner: Đỗ Minh Phương | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

17.4 Preconditions

17.4.1 Required System and Actor State

Administrator has an active Administrator session. Two approved fixed layout templates, seated and sleeper, are available.

17.5 Postconditions

17.5.1 Success

A valid complete vehicle/layout record is created, or an explicitly permitted maintenance action is saved.

17.5.2 Failure and Minimal Guarantees

Invalid changes do not modify a vehicle, break seat references or alter existing Bookings.

17.6 Extension Points

17.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

18. UC-A02 Manage Routes and Stops

18.1 Manage Routes and Stops

18.1.1 Brief Description

Allow the Administrator to create and maintain routes and their ordered stops.

Related Feature(s): F15

Primary Actor: Administrator

Supporting Actor(s): None

Trigger: The Administrator requests to create a route or maintain an existing route and its stops.

18.2 Flow of Events

18.2.1 Basic Flow

1. Administrator opens Route/Stop management.

2. The system verifies access and lists Routes and Stops.

3. Administrator chooses to create a Route and enters a unique code and name.

4. Administrator selects or creates Stops with names and addresses.

5. Administrator arranges at least two distinct Stops in travel order.

6. The system validates required Route/Stop fields, unique Route code, Stop count/order and absence of repeated Stops.

7. The system saves the Route and ordered Stop sequence.

8. The system displays the saved Route and confirms success.

18.2.2 Alternative Flows

18.2.2.1 A1 — Maintain existing Route or Stop

A1.1 At Step 3 or 4, Administrator selects an existing record.

A1.2 The system loads the selected Route/Stop. Administrator edits only fields that preserve used journey meaning; a Route's Stop structure/order is immutable after assignment to any Trip. Structural rerouting requires a new Route.

A1.3 The system validates the edited fields under Step 6 rules, then Administrator reviews and explicitly confirms the update.

A1.4 The system updates the selected record without changing its identity or dependent journey references, displays the saved result and ends this alternative; it does not create another Route.

18.2.2.2 A2 — Invalid Route/Stop information

A2.1 At Step 6, a required field is missing, a Route code conflicts, fewer than two Stops exist, a Stop repeats or its order is invalid.

A2.2 The system requests corrections and does not save an invalid sequence.

18.2.2.3 A3 — Delete or deactivate

A3.1 At Step 3 or 4, Administrator requests removal.

A3.2 The system checks references and displays the consequence. An unused unreferenced record may be deleted only after Administrator confirms; the system removes it and confirms completion.

A3.3 A Stop used by a Route or Route used by a Trip cannot be deleted. Administrator may explicitly confirm retirement from new assignments; the system saves the availability change while preserving references and displays the result. Declining confirmation changes nothing.

A3.4 The alternative ends after the saved result; unauthorized, missing-record or save failures follow A4.

18.2.2.4 A4 — Unauthorized, missing record or save failure

A4.1 At Step 2, 3, 4 or 7, permission is invalid, the selected record is missing or saving fails.

A4.2 The system denies access or reports the error; no partial Route sequence is treated as successfully saved.

18.3 Special Requirements

18.3.1 Business Rules

[BR-A02-01] A Route requires a unique code, name and ordered list of at least two distinct Stops; each Stop requires a name and address. The same Stop may not repeat within one Route.

[BR-A02-02] Once a Route is assigned to a Trip, its Stop structure/order is protected. Changed routing uses a new Route record; existing Trip and Booking meaning is preserved.

[BR-A02-03] Deletion is allowed only before use and only without dependent records. Used Routes/Stops remain available for historical references and may be retired from new assignment.

[BR-A02-04] All Segment availability and fare calculations follow this saved Stop order.

18.3.2 Approved Decisions and Scope

Minimum two distinct Stops, no repeated Stop, required Route/Stop fields, protected used Routes and deletion only before use are approved.

Specification owner: Đỗ Minh Phương | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

18.4 Preconditions

18.4.1 Required System and Actor State

Administrator has an active Administrator session and can manage the operator's Route and Stop records.

18.5 Postconditions

18.5.1 Success

A valid Route with complete ordered Stops or an allowed maintenance action is saved.

18.5.2 Failure and Minimal Guarantees

Invalid/unauthorized/failed actions do not change the Stop order of an existing sold journey or delete dependent history.

18.6 Extension Points

18.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

19. UC-A03 Manage Schedules and Trip Instances

19.1 Manage Schedules and Trip Instances

19.1.1 Brief Description

Allow the Administrator to create and maintain departure schedules and specific trip instances for the operator's routes.

Related Feature(s): F15

Primary Actor: Administrator

Supporting Actor(s): None

Trigger: The Administrator requests to create a departure schedule or maintain an existing schedule or trip instance.

19.2 Flow of Events

19.2.1 Basic Flow

1. Administrator opens schedule and Trip management.

2. The system verifies access and lists departure schedules and Trip instances.

3. Administrator creates a schedule with a Route, operating weekdays/dates and planned first-Stop departure time.

4. Administrator selects a date range of at most 30 days, assigns a vehicle, and supplies planned times at all Stops plus nonnegative fares for consecutive legs.

5. The system validates Route/vehicle availability, increasing Stop times, the date range, complete leg fares, duplicate Trip generation and vehicle time conflicts, then displays a preview of the concrete Trips.

6. Administrator reviews and explicitly confirms the preview.

7. The system creates the confirmed Trip instances with their vehicle/layout, planned Stop times and fare information associated with the schedule and Route.

8. The system displays the saved schedule/Trips and confirms success.

19.2.2 Alternative Flows

19.2.2.1 A1 — Maintain a schedule

A1.1 At Step 3, Administrator selects an existing schedule.

A1.2 The system loads the selected schedule. Administrator edits its Route, operating days/dates or first-Stop departure time; the system validates the schedule fields and displays the proposed revision.

A1.3 Administrator explicitly confirms. The system saves an update to that schedule identity, leaves already generated Trips unchanged and displays the saved revision.

A1.4 If Administrator also requests later generation, continue at Step 4 with a bounded date range and a new preview. Otherwise this alternative ends with only the schedule updated.

19.2.2.2 A2 — Maintain a concrete Trip

A2.1 At Step 3, Administrator selects a Scheduled Trip.

A2.2 The system loads the selected Scheduled Trip with its identity and current allocation state. Administrator changes permitted vehicle, Route, times or fares.

A2.3 The system applies Step 5 validation to an UPDATE: the Trip being edited is excluded from self-conflict/duplicate detection, but all other Trip conflicts and protected-allocation rules still apply.

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A2.4 Administrator reviews and explicitly confirms the update. Immediately before saving, the system rechecks no Confirmed Booking or effective Seat Hold exists; a new allocation follows A4.

A2.5 The system updates that same Trip identity, preserves historical Booking/Ticket facts, displays the updated Trip and ends this alternative. It does not execute the new-Trip creation at Step 7.

19.2.2.3 A3 — Invalid, conflicting or missing input

A3.1 At Step 5, a Route/vehicle is missing, times are not ordered, fares are missing/negative, a range exceeds 30 days, a Trip would duplicate an existing one or vehicle operating intervals overlap.

A3.2 The system identifies the issue and requests correction; the preview cannot be confirmed while invalid.

19.2.2.4 A4 — Trip already has an effective allocation

A4.1 At Step 5 or 7, a requested structural/time/fare edit conflicts with a Confirmed Booking or unexpired hold, including one created after preview.

A4.2 The system refuses the edit and preserves sold/held journey information. Cancellation, if required while Scheduled, follows UC-S04.

19.2.2.5 A5 — Delete or retire

A5.1 At Step 3, Administrator requests removal.

A5.2 The system allows deletion only of an unreferenced schedule/Trip. Used records remain in history; a schedule may stop future generation, and a Scheduled Trip with Bookings must use the cancellation process instead of deletion.

19.2.2.6 A6 — Unauthorized access or save failure

A6.1 At Step 2 or 7, permission is invalid or saving cannot complete consistently.

A6.2 The system denies access or reports failure without partial Trip creation or false success.

19.3 Special Requirements

19.3.1 Business Rules

[BR-A03-01] A departure schedule defines Route, operating days/dates and first-Stop departure time. Concrete Trips are generated only after an Administrator previews and confirms a date range no longer than 30 days; no indefinite background generation is required.

[BR-A03-02] Each Trip has an assigned vehicle/layout, Route, planned times for every Stop and a nonnegative fare for every consecutive leg. A Segment fare is the sum of its leg fares, equal across seats on that Trip.

[BR-A03-03] Planned Stop times must follow Stop order and valid travel dates. One vehicle cannot have overlapping Trip intervals from first departure to final arrival, and repeated generation must not create duplicate Trip instances.

[BR-A03-04] Do not change a Trip's vehicle, Route, times or fares while it has a Confirmed Booking or effective hold; validate again when saving. Existing Booking/Ticket journey and fare history is retained.

[BR-A03-05] Schedule changes affect later generation only. Already generated Trips do not silently change with the schedule.

[BR-A03-06] Delete only unreferenced schedules/Trips. Preserve used history; stop future generation or use the approved Trip-cancellation process where applicable.

19.3.2 Approved Decisions and Scope

Vehicle assignment belongs to A03. Preview-based generation for up to 30 days, per-Stop times, per-leg fares, conflict validation, protected allocated Trips and non-retroactive schedule changes are approved.

Specification owner: Đỗ Minh Phương | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

19.4 Preconditions

19.4.1 Required System and Actor State

Administrator has an active Administrator session. At least one valid Route and available vehicle/layout can be selected.

19.5 Postconditions

19.5.1 Success

The confirmed schedule/Trip records are saved with complete vehicle, Stop-time and fare data, or permitted maintenance is applied without altering protected allocations.

19.5.2 Failure and Minimal Guarantees

Invalid/conflicting/unauthorized/failed operations create no duplicate or partial Trips and do not change sold/held journey information.

19.6 Extension Points

19.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

20. UC-A04 Review Bookings and Operations Overview

20.1 Review Bookings and Operations Overview

20.1.1 Brief Description

Allow an authorized Administrator to review the bus operator's bookings, locate relevant records, and inspect basic booking and operational statuses.

Related Feature(s): F16 — Booking Management / Operations Overview

Primary Actor: Administrator

Supporting Actor(s): None

Trigger: The Administrator opens the booking management / operations overview.

20.2 Flow of Events

20.2.1 Basic Flow

1. Administrator opens the Booking/operations overview.

2. The system verifies the session and administrative review permission.

3. The system retrieves authorized Booking, Ticket, Payment, reconciliation and Refund records for the operator.

4. The system displays Booking count, Ticket counts by Valid/Checked In/Cancelled status, total verified successful Payment amount and the count of transactions requiring reconciliation.

5. Administrator selects filters for Booking reference, Route/Trip, travel date, Booking status or Payment status.

6. The system validates the filters and displays matching Bookings with permitted contact and journey information.

7. Administrator selects a Booking.

8. The system displays its passenger/contact details needed for operations, individual Tickets, Segment/seats, attempt-level Payment results and separate reconciliation/Refund statuses.

9. Administrator returns to the list without changing a Booking, seat, Payment, Ticket or Refund.

20.2.2 Alternative Flows

20.2.2.1 A1 — Unauthorized access

A1.1 At Step 2, the session is missing, expired or not an authorized Administrator session.

A1.2 The system denies access without exposing protected records.

20.2.2.2 A2 — No records or no matches

A2.1 At Step 3 or 6, the dataset is empty or no permitted Booking matches.

A2.2 The system displays a clear empty result and allows revised filters; no data changes.

20.2.2.3 A3 — Invalid filters

A3.1 At Step 6, a filter value or date is invalid.

A3.2 The system requests correction at Step 5 and does not display an incorrectly filtered result.

20.2.2.4 A4 — Record changed, inaccessible or unavailable

A4.1 At Step 3 or 8, records cannot be loaded or the selected record has changed.

A4.2 The system refreshes current authorized facts when possible or reports retrieval failure; it does not invent statuses.

20.2.2.5 A5 — Modification requested

A5.1 At Step 8, an administrative request attempts to edit/cancel a passenger Booking, change a seat, approve a payment or execute a refund through this overview.

A5.2 The system does not provide those modifying operations; this use case remains read-only. Reconciliation/refund work occurs in the operator's external back-office process.

20.3 Special Requirements

20.3.1 Business Rules

[BR-A04-01] F16 is read-only authorized review, search/filter and operational overview for one operator. Booking modification, seat changes, manual payment approval and refund execution are outside UC-A04 and the current application scope.

[BR-A04-02] Approved filters are Booking reference, Route/Trip, travel date, Booking status and Payment status. Necessary passenger/contact name, email and telephone may be shown to Administrator, with journey, seat and lifecycle records; passwords/authentication data are never shown. The Booking travel-date filter uses its selected boarding Stop's local calendar date under BR-P04-05; a Trip's operational departure date is separately labelled if shown.

[BR-A04-03] The overview reports Booking count; Ticket counts by Valid, Checked In and Cancelled; total provider-verified Succeeded Payment amount; and transactions with Review Required reconciliation. Valid plus Checked In constitutes the current confirmed-issued Ticket count.

[BR-A04-04] The successful-Payment total is money recorded, not revenue or profit; unmatched/extra successful amounts remain visible with separate reconciliation/Refund status. Label mock demonstration figures as simulated rather than provider-verified.

[BR-A04-05] Use the shared status sets: Booking Pending/Confirmed/Expired/Cancelled; Payment Pending/Succeeded/Failed/Cancelled; reconciliation Not Required/Review Required/Resolved; Ticket

Valid/Checked In/Cancelled; Refund Not Required/Pending/Completed/Failed.

[BR-A04-06] Only verified eligible payment can issue Tickets in P09. UC-A04 never overrides hold expiry, ownership, cancellation or Ticket-validity rules.

20.3.2 Approved Decisions and Scope

Read-only F16, explicit filters/visible fields, the limited summary measures and shared lifecycle labels are approved. Reconciliation and refund execution remain outside this UI.

Specification owner: Liên Trung Hiếu | Requirements baseline: 1.5 | Business decisions confirmed: 23 October 2026; flow review: 24 October 2026.

20.4 Preconditions

20.4.1 Required System and Actor State

Administrator has an active Administrator session and is authorized to review the single operator's Booking information; records may be empty.

20.5 Postconditions

20.5.1 Success

Administrator reviews authorized operational records, filters and summaries; all journey/payment/refund states remain unchanged.

20.5.2 Failure and Minimal Guarantees

Unauthorized or failed access discloses no protected data and never modifies records or creates a false payment/refund confirmation.

20.6 Extension Points

20.6.1 None Defined

No UML extension point is defined in this baseline. Alternative flows and references to other use cases do not imply an «extend» relationship.

APPENDIX A — UC-C01 Validate Ticket for Check-in

A.1 Purpose and inclusion — Supporting behavior included by UC-S03, with Staff as the actor of the including use case. It factors existing validation; it adds no permission, lifecycle state or user goal.

A.2 Inputs and normal entry — selected Trip reference, individual Ticket reference, recorded boarding Stop context and Staff session. Missing/inaccessible references are handled as failures; no assumption of eligibility is made.

A.3 Basic Flow

1. Verify active Staff session and check-in permission; otherwise return Unauthorized without disclosure.

2. Retrieve current Ticket, its Booking and selected Trip; missing or unreadable facts return NotFound/Unavailable without guessing.

3. Verify Ticket belongs to the selected Trip, Booking is Confirmed and Ticket is Valid. Already Checked In returns AlreadyCheckedIn; cancelled/unconfirmed records return InvalidJourney.

4. Verify Trip is Scheduled, or Departed with an intermediate boarding Stop. Reject Cancelled/Completed Trips and first-Stop boarding after departure. On the first invocation, return the recorded boarding Stop for Staff review in S03 Step 5. On the pre-save invocation after Staff confirmation, also validate the reviewed boarding Stop against the current recorded Stop.

5. Return Eligible with current Ticket/Booking/Trip/Stop references and traveller/seat facts permitted for operational review. S03 must invoke these checks again immediately before writing check-in.

A.4 Alternatives — Unauthorized→S03 A1; missing/wrong Trip or Stop→A2; already checked in→A3; cancelled/unconfirmed/inactive→A4; retrieval failure→A6. Every failure returns control to S03; it never writes check-in or changes any journey record.

A.5 Rules — [BR-C01-01] Apply BR-S03-01 through BR-S03-05 as the authoritative per-Ticket, boarding, confirmation/audit and no-reversal rules. [BR-C01-02] An Eligible result is informational until S03 revalidates and saves consistently; C01 cannot reserve a seat, issue a Ticket or alter a Payment.

A.6 Postconditions — Eligible or one explicit failure reason is returned. Ticket, Booking, Trip, Payment, Refund and allocations remain unchanged.

A.7 Extension Points — None. The include arrow runs from UC-S03 to UC-C01.

APPENDIX B — Business-rule reference convention

BR-<UC>-<number> identifies each rule paragraph at its authoritative UC section; use that ID in PA3 data constraints, interaction designs and test cases. IDs are references, not additional requirements. Shared behavior refers to the original rule ID rather than copying a competing policy.

Model policy — UC-P08 Create Temporary Booking ends with Pending + 10-minute hold and no Ticket. P09 is independently initiated for an owned payable Booking. There is no P08→P09 include/extend. UC-P10 extends P06 only at EP-P06-CANCEL; every alternative flow is not automatically an extend.

Revision integrity — 1.0–1.2 remain October history; 1.3 is the feedback correction dated 4 November 2026. Artifact completion is not member sign-off or implemented test evidence. Baseline 1.4 adds the second editorial quality pass, note-free views and acceptance/traceability designs; DEC-01/02 were subsequently approved as option A/A on 4 November 2026 and incorporated in baseline 1.5.

Companion documents — Team 11 — Software Development Plan 1.7 and Team 11 — Vision 1.7 are the consolidated PA2 successors to their PA1 versions. Their organization, operating assumptions and competitor analysis do not change this UC 1.5 policy baseline or model 1.5. Sprint 2 Backlog retains 33 tasks and 68.5 planned hours. W1, W3 and the 24 October Submission Status retain separate reporting periods. This is an editorial reference update dated 4 November 2026; it adds no UC, BR or test outcome.

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APPENDIX C — USE-CASE MODEL — DETAILED VIEWS

Requirements baseline 1.5 | Visual edition 2.1 | 6 November 2026

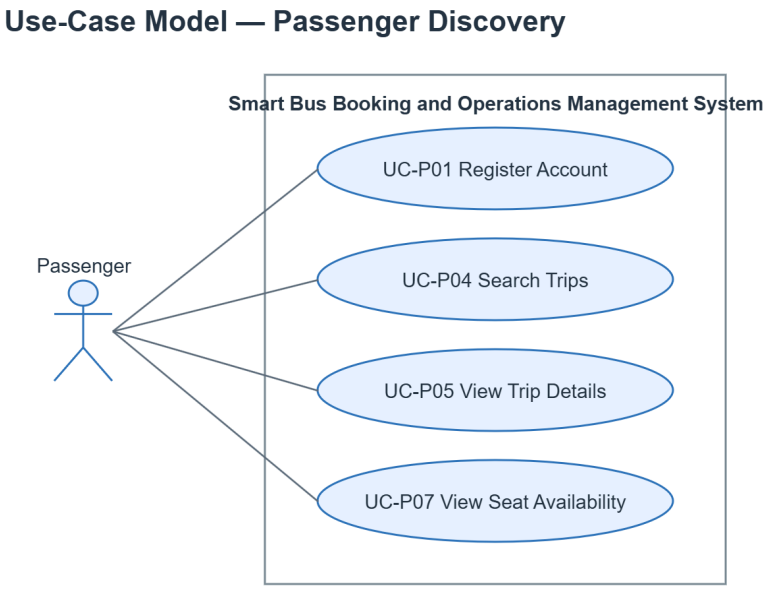
[REF-01] PA3_Group11_UseCaseModel_v2.1.drawio — editable six-view UML model; the same source is used in SAD section 3.

[REF-02] PA3 UC Model v2.1 — six PNG/SVG exports: overview, discovery, booking, AI, operations and shared authentication. See the source register in Appendix D.

The diagrams contain only actors, use cases, system boundaries, UML relationship labels/guards and the P06 extension-point compartment. Explanations belong in the paragraphs below and Appendix D — Acceptance & Traceability. All views refer to the same model; repeated actor symbols denote the same role. No new user goal is added by supporting C01.

Passenger discovery

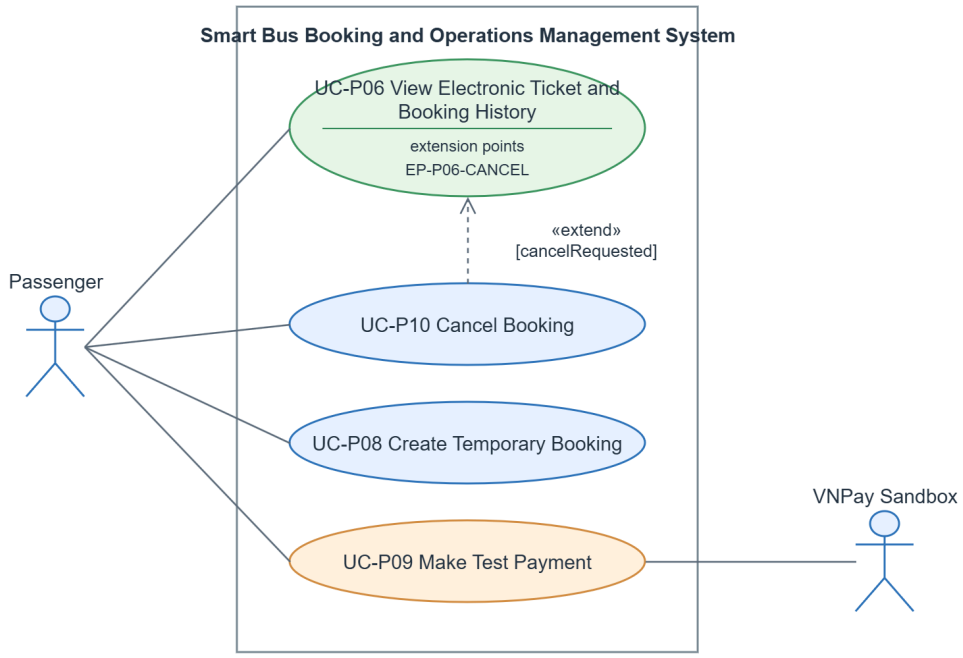
P01 is public registration without automatic login. P04/P05/P07 are public search/details/availability. Public viewing creates no hold; P08 requires Passenger authentication and a fresh availability check.



Booking, payment and cancellation

P08 ends at one Pending Booking with a fixed ten-minute hold. P09 is a separately triggered payment goal. P10 extends P06 at EP-P06-CANCEL only when cancelRequested is true: Passenger chooses Cancel from the owned Booking details. Eligibility/confirmation are handled by P10; viewing is complete without cancellation.

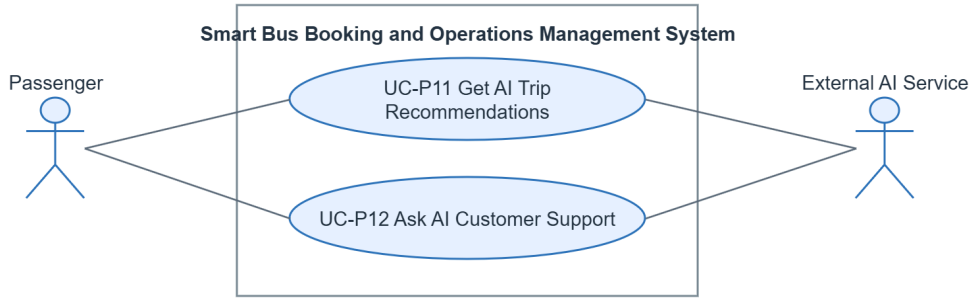
Use-Case Model — Booking, Payment and Cancellation



AI assistance

External AI Service (Google Gemini Developer API) supports P11/P12. P11 validates filters and retrieves real Trip records; P12 uses approved policies and authorized facts. Private Booking support requires Passenger ownership. No AI action creates/reserves/pays/cancels a journey.

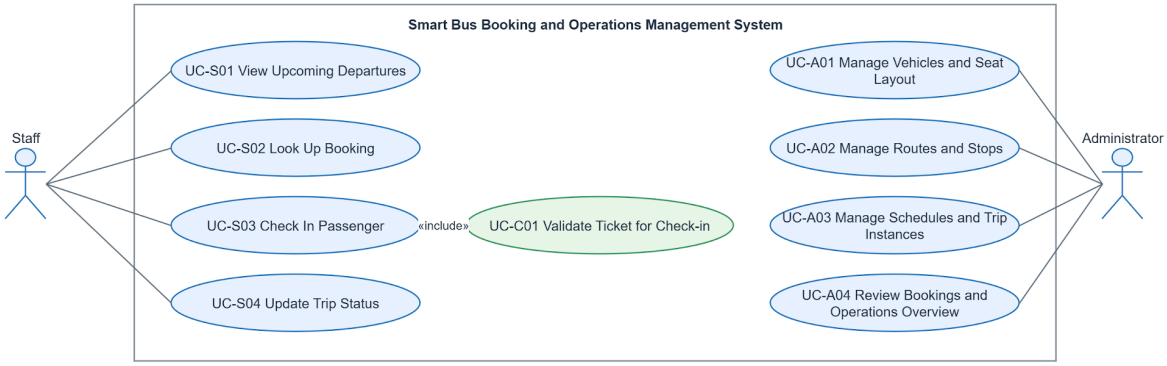
Use-Case Model — AI Assistance



Staff and administrator operations

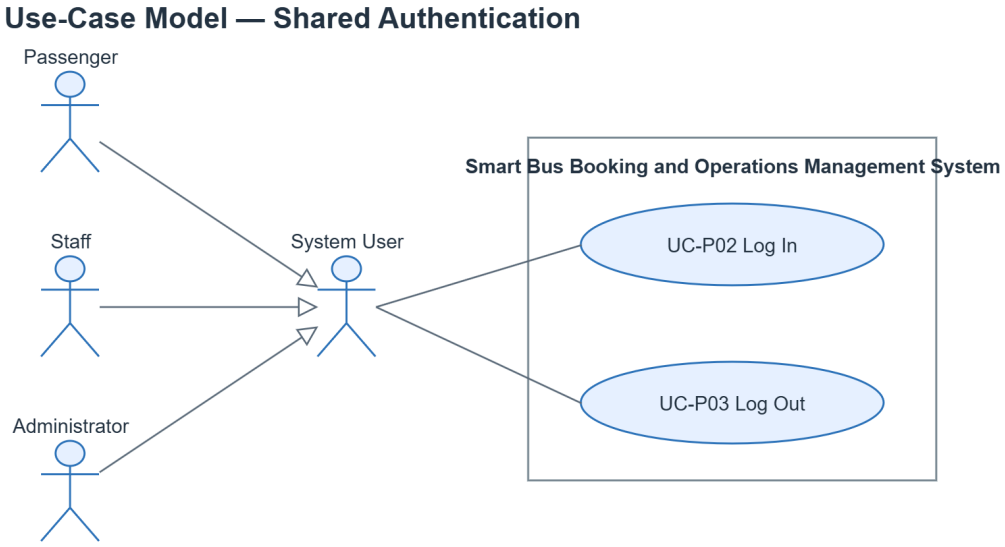
S03 includes C01 before Staff review and invokes it again before writing check-in. C01 is side-effect-free supporting behavior and adds no user goal. A04 remains read-only. Operational permissions are listed in the access-control matrix, not inferred from the diagram placement.

Use-Case Model — Staff and Administrator



Shared authentication

The hollow-triangle generalizations point from Passenger/Staff/Administrator to System User; they share its P02/P03 associations. This does not make the three roles interchangeable. A protected UC checks an existing session rather than including interactive Login on every invocation.



APPENDIX D — ACCEPTANCE & TRACEABILITY

UC baseline 1.5 | 4 November 2026. This tab specifies acceptance designs, not executed results. It covers all 20 user-goal UCs, supporting C01, 96 primary BR paragraphs and NFR01–NFR09. DEC-01 and DEC-02 are approved as option A/A by the project lead. TC-09/TC-25 now have finalized expected outcomes; all cases still require implementation-time execution evidence.

Evidence states — All 62 acceptance designs have defined inputs/actions/expected outcomes and are ready for implementation-time execution. DEC-01/02 were approved A/A by the lead; no case is blocked by those decisions. Designed is not Passed: actual results, environment, records and reviewer evidence must be recorded when the software is available. PA3 design IDs are handoff targets rather than implemented API/design claims.

D.1. UC / BR / design / acceptance traceability

UC / Feature	Authoritative BR IDs	PA3 design target	Acceptance cases
UC-P01 / F01	BR-P01-01 to BR-P01-03	D-AUTH / account creation	TC-01, TC-02
UC-P02 / F01	BR-P02-01 to BR-P02-04	D-AUTH / sessions	TC-03, TC-04, TC-05
UC-P03 / F01	BR-P03-01 to BR-P03-03	D-AUTH / session revocation	TC-06
UC-P04 / F02	BR-P04-01 to BR-P04-05	D-SEARCH / TripStop queries	TC-07, TC-08, TC-09
UC-P05 / F04	BR-P05-01 to BR-P05-03	D-SEARCH / details/fare	TC-10
UC-P06 / F08	BR-P06-01 to BR-P06-05	D-TICKET / ownership/history	TC-11, TC-12
UC-P07 / F05	BR-P07-01 to BR-P07-04	D-INVENTORY / Segment overlap	TC-13, TC-14
UC-P08 / F06	BR-P08-01 to BR-P08-05	D-BOOKING / atomic hold	TC-15, TC-16, TC-17
UC-P09 / F07	BR-P09-01 to BR-P09-06	D-PAYMENT / verification/idempotency	TC-18, TC-19, TC-20, TC-21, TC-22
UC-P10 / F09	BR-P10-01 to BR-P10-07	D-CANCEL / refund separation	TC-23, TC-24, TC-25, TC-26
UC-P11 / F03	BR-P11-01 to BR-P11-07; shared BR-P04-05	D-AI / validated search filters	TC-27, TC-28, TC-29, TC-30; shared date case TC-09
UC-P12 / F10	BR-P12-01 to BR-P12-06	D-AI / authorized factual support	TC-31, TC-32, TC-33
UC-S01 / F11	BR-S01-01 to BR-S01-03	D-STAFF / departure projection	TC-34
UC-S02 / F11	BR-S02-01 to BR-S02-04	D-STAFF / authorized lookup	TC-35, TC-36

UC / Feature	Authoritative BR IDs	PA3 design target	Acceptance cases
UC-S03 / F12	BR-S03-01 to BR-S03-05	D-CHECKIN / validation and atomic save	TC-37, TC-38, TC-39
UC-S04 / F13	BR-S04-01 to BR-S04-06	D-OPS / Trip lifecycle	TC-40, TC-41, TC-42
UC-A01 / F14	BR-A01-01 to BR-A01-04	D-OPS / vehicle protection	TC-43, TC-44
UC-A02 / F15	BR-A02-01 to BR-A02-04	D-OPS / ordered Route	TC-45, TC-46
UC-A03 / F15	BR-A03-01 to BR-A03-06	D-OPS / generation/conflict guard	TC-47, TC-48, TC-49, TC-50
UC-A04 / F16	BR-A04-01 to BR-A04-06	D-OVERVIEW / read-only projection	TC-51, TC-52
UC-C01 / supports F12	BR-C01-01 / BR-C01-02; refer BR-S03-01–05	D-CHECKIN / side-effect-free validation	TC-37, TC-38, TC-39, TC-53

D.2. Access-control matrix

Public means no private or operational authority. Ownership is rechecked server-side on every protected request; an Administrator is not implicitly a Passenger. P02 starts a session only from unauthenticated entry. System User generalization in the diagrams shares P02/P03 associations; it grants no operational permissions.

Operation	Anonymous	Passenger	Staff	Administrator	Scope / condition
UC-P01	Allow	No active session	No active session	No active session	Public entry creates Passenger only; normal entry requires no authenticated session.
UC-P02	Allow	After logout	After logout	After logout	Active account / credential and restriction checks; one assigned role.
UC-P03	No session	Allow	Allow	Allow	Revoke current session; no Booking/hold lifecycle change.
UC-P04	Public	Public	Public	Public	No allocation/private records; protected follow-up uses role/ownership checks.
UC-P05	Public	Public	Public	Public	No allocation/private records; protected follow-up uses role/ownership checks.
UC-P06	Deny	Own only	Deny	Deny	P06/P08/P09/P10: authenticated Passenger; P08 creates an owned Booking, other actions enforce record ownership.
UC-P07	Public	Public	Public	Public	No allocation/private records; protected follow-up uses

Operation	Anonymous	Passenger	Staff	Administrator	Scope / condition
					role/ownership checks.
UC-P08	Deny	Own only	Deny	Deny	P06/P08/P09/P10: authenticated Passenger; P08 creates an owned Booking, other actions enforce record ownership.
UC-P09	Deny	Own only	Deny	Deny	P06/P08/P09/P10: authenticated Passenger; P08 creates an owned Booking, other actions enforce record ownership.
UC-P10	Deny	Own only	Deny	Deny	P06/P08/P09/P10: authenticated Passenger; P08 creates an owned Booking, other actions enforce record ownership.
UC-P11	Public	Public	Public	Public	No allocation/private records; protected follow-up uses role/ownership checks.
UC-P12	Public FAQ/Trip	Public + own Booking	Public FAQ/Trip only	Public FAQ/Trip only	Private Booking branch requires Passenger role + ownership. No Staff/Admin-specific AI support.
UC-S01	Deny	Deny	Allow	Deny	Single-operator operational records; S02 limited fields; S03 per-Ticket validation, S04 transition guards.
UC-S02	Deny	Deny	Allow	Deny	Single-operator operational records; S02 limited fields; S03 per-Ticket validation, S04 transition guards.
UC-S03	Deny	Deny	Allow	Deny	Single-operator operational records; S02 limited fields; S03 per-Ticket validation, S04 transition guards.
UC-S04	Deny	Deny	Allow	Deny	Single-operator operational records; S02 limited fields; S03 per-Ticket validation, S04 transition guards.
UC-A01	Deny	Deny	Deny	Allow	A01–A03 protected maintenance; A04 read-only. No account/FAQ manager or manual payment/refund override.
UC-A02	Deny	Deny	Deny	Allow	A01–A03 protected maintenance; A04 read-only. No account/FAQ manager or manual payment/refund override.
UC-A03	Deny	Deny	Deny	Allow	A01–A03 protected maintenance; A04 read-only. No account/FAQ manager or manual payment/refund override.
UC-A04	Deny	Deny	Deny	Allow	A01–A03 protected maintenance; A04 read-only. No account/FAQ

Operation	Anonymous	Passenger	Staff	Administrator	Scope / condition
					manager or manual payment/refund override.
C01 direct entry	Deny	Deny	Via S03 only	Deny	Included internal validation, not a separate public modifying endpoint.
P09 provider callback	No user authority	No user authority	No user authority	No user authority	Server-side provider authenticity/identity/amount verification; no browser redirect or admin override substitutes for it.

D.3. Lifecycle invariants

Trigger	Booking / allocation	Ticket	Payment / refund / invariant
P08 succeeds	Pending + fixed ten-minute hold	None issued	Saved fare and contact/traveller snapshot; P09 starts separately.
Eligible verified P09 success	Confirmed; hold becomes confirmed allocation	Exactly one Valid Ticket per paid seat	First eligible verified success only; duplicate callbacks are idempotent.
Hold expires	Expired; no effective hold remains	No issuance from expired hold	Attempt states stay factual; late money goes to reconciliation, never resurrects a seat.
Passenger cancels Pending	Cancelled; only its holds released	No usable Ticket	Unpaid: Refund Not Required. Later verified money follows P09 unmatched repayment.
Eligible Passenger cancels Confirmed	Whole Booking Cancelled; only its Segment released	All its Tickets Cancelled	One 90% Refund Pending; no Checked In Ticket, ≥12 hours before selected boarding. Whole-Booking rounding follows approved BR-P10-07; charge is the paid amount minus rounded refund.
Operator cancels Scheduled Trip	Affected Pending/Confirmed Bookings cancelled; release holds/allocations	Invalidate all affected, including Checked In	100% eligible paid refund request; reconcile prior refunds and never repay above paid amount.
Late / extra verified money	No new confirmation or lost allocation restored	No additional Ticket	Payment Succeeded + Review Required; full unmatched repayment is external.
Staff confirms eligible check-in	Confirmed Booking/allocation unchanged	Only selected Ticket becomes Checked In	C01 rechecked before save; Payment unchanged; no seat release or duplicate check-in.
A04 / AI / public viewing	No journey mutation	No journey mutation	A04 is read-only; AI cannot reserve/pay/cancel/issue/execute refunds.

D.4. Acceptance case designs

Common fixture: one operator, Route A→B→C with increasing Asia/Ho_Chi_Minh Stop timestamps, vehicle seats S1–S6, Passenger X/Y, Staff S and Administrator A. Default A→B fare 100,000 VND and B→C 150,000; use Trip dates wholly within one day unless a case says otherwise. Boundary/concurrency tests use a controlled authoritative clock. Every case remains Designed — not executed. DEC-01/02 are approved A/A; TC-09/TC-25 expectations are finalized. For overnight searches, set the authoritative clock before the common first-Stop cutoff. Capture input, before/after records, actor, timestamps, actual output and evidence at execution.

Case / references	Preconditions and action	Expected outcome
TC-01 BR-P01-01, BR-P01-02, BR-P01-03	Submit a new Passenger with all five required inputs, unused email and matching password/confirmation. Then attempt a second registration with the same email.	First creates exactly one Passenger account and no session; password is not returned/stored as plaintext. Duplicate creates no account and leaves the original unchanged. Public input cannot create Staff/Admin.
TC-02 BR-P01-01	Repeat registration with each mandatory input missing, mismatched confirmation and invalid email.	Each invalid submission identifies the correction; no partial usable account or authenticated session exists.
TC-03 BR-P02-01, BR-P02-03	Sign in separately as active Passenger, Staff and Administrator, then with an inactive account and incorrect password.	Active accounts receive their assigned role only. Inactive/incorrect credentials create no session or protected disclosure; no permanent lock is introduced.
TC-04 BR-P02-02, BR-P02-03	Five consecutive incorrect attempts within 15 minutes; retry during the next five-minute restriction, after it, and after a successful login.	Restriction lasts five minutes; restricted attempts create no session. Login becomes available after restriction. Success resets the consecutive-failure counter; account is not permanently locked.
TC-05 BR-P02-04	Idle Passenger for 29:59 then 30:00; Staff/Admin for 14:59 then 15:00. Request protected data at each boundary.	Before timeout, otherwise valid access may proceed. At or after role timeout, protected requests are denied; session activity timing is server-authoritative.
TC-06 BR-P03-01, BR-P03-02, BR-P03-03	For each role, log out and replay its previous protected request. For Passenger, keep a Pending Booking with a hold created at t0.	Old session cannot access protected actions. Logout does not cancel the Booking or change hold expiry t0+10 minutes; no new hold or allocation is created.
TC-07 BR-P04-01, BR-P04-02	Public search with matching Scheduled, Departed, Completed and Cancelled Trips. Test first-Stop departure minus 30 minutes and one second before it, including an intermediate boarding Stop.	Only Scheduled Trips still before the common cutoff are sellable. Exactly at cutoff reject new booking; intermediate boarding does not move that cutoff. Search creates no allocation.
TC-08 BR-P04-03, BR-P04-04	A→B costs 100,000 VND and B→C 150,000; search A→C and B→C. Modify optional descriptions, not saved Trip facts.	A→C displays 250,000 and B→C 150,000 for every seat of that Trip; times/prices match saved Trip records and results are sorted by planned boarding time.
TC-09 BR-P04-05	Trip first Stop A departs 10 November 23:00; B boarding is 11 November 01:00. Search B→C for 11 November.	Using option A, an advance B→C search for 11 November includes the Trip while sales remain open; a search for 10 November excludes it for that boarding Stop. Convert UTC Stop timestamps to Asia/Ho_Chi_Minh before extracting the date. Exactly at the first-Stop cutoff, reject sale regardless of later boarding. Ordinary and AI filters agree.
TC-10	Open a Trip publicly; omit optional	Optional gaps are labelled but details remain

Case / references	Preconditions and action	Expected outcome
BR-P05-01, BR-P05-02, BR-P05-03	description, then separately remove essential Stop times/layout/leg fare data. Compare Segment fare with TC-08.	viewable. Missing essential data disables booking. No invented facts; Segment fare remains the sum of leg fares, equal across seats.
TC-11 BR-P06-01, BR-P06-02, BR-P06-03	Passenger X views a two-seat paid Booking, an empty history and a Booking owned by Y.	X sees exactly two unique individual Ticket codes with traveller/Trip/seat/Segment/time/fare/status fields. Empty history is distinct from error. Y's facts and record existence are not disclosed.
TC-12 BR-P06-04, BR-P06-05	Display Pending, Confirmed, Expired and Cancelled Bookings, with failed/late/extra successful Payments and cancelled/checked-in Tickets.	Booking, attempt Payment, reconciliation, Ticket and Refund states are displayed separately using the approved sets. Payment Succeeded alone never produces a usable Ticket.
TC-13 BR-P07-02	Same Trip/seat: one effective A→B allocation; query B→C, A→C and A→B. Repeat with Confirmed Booking, live hold, expired hold and Cancelled Booking.	B→C can reuse the seat. Overlapping live holds/Confirmed allocations block it. Expired/cancelled records do not block it; no two effective overlapping allocations are introduced.
TC-14 BR-P07-01, BR-P07-03, BR-P07-04	Create a hold at t0; refresh the map at t0+9:59, t0+10:00 and after expiry. Public viewer then attempts selection using a stale available map.	Expiry remains t0+10:00 and the display updates. Public viewing reserves nothing. New booking rechecks authoritative availability; stale rendering never grants a seat.
TC-15 BR-P08-01	Create bookings with 1 and 6 distinct seats and traveller names. Try 0/7 seats, mixed Trips/Segments, missing traveller name and unauthenticated creation.	Valid requests create one Passenger-owned Booking for one Trip/Segment. Invalid/unauthenticated requests create no Booking or partial holds; protected creation requires Passenger role.
TC-16 BR-P08-02, BR-P08-03, BR-P08-04	For an open Trip, confirm two seats at 250,000 each; refresh/retry and request payment later. Also submit a new booking at the common cutoff.	Successful creation records Pending, saved total 500,000 and a fixed ten-minute hold; no Ticket or automatic payment. Refresh/retry cannot extend expiry or mutate the confirmed fare. At cutoff, creation is rejected.
TC-17 BR-P08-05	Two simultaneous users request the same Trip/seat on overlapping Segments; also inject failure while creating a multi-seat hold.	At most one conflicting request succeeds. Failure rolls back the entire attempted Booking/hold set; no partial effective allocation or false success survives. Retry does not violate the invariant.
TC-18 BR-P09-01, BR-P09-02	Initiate one attempt, leave outcome unknown, then request a parallel retry. After terminal failures, initiate up to three attempts and request a fourth.	References are unique, one Pending maximum and three initiated attempts maximum. Unknown stays Pending, not Failed; no parallel attempt is permitted and the original hold/amount never changes.
TC-19 BR-P09-03, BR-P09-04	Before expiry, receive the first provider-verified exact-amount success for a two-seat Pending Booking. Replay the same callback repeatedly.	Booking becomes Confirmed once; exactly two unique Valid Tickets are issued. Duplicate notifications cause no second confirmation/allocation/Ticket. Payment and reconciliation remain distinct.
TC-20 BR-P09-04	Submit invalid signature, wrong Booking/attempt reference, wrong amount and a browser success redirect without authoritative server verification.	No verified success, Booking confirmation or Ticket issuance. Unavailable/inconsistent outcomes remain unconfirmed and follow the Review Required rule rather than being treated as eligible payment.
TC-21 BR-P09-03,	Deliver verified success after expiry/cancellation, then a distinct extra	Record actual Payment Succeeded and Review Required; late/cancelled payment creates no Ticket

Case / references	Preconditions and action	Expected outcome
BR-P09-05	success after an earlier accepted success.	or recovered seat. Extra money does not duplicate existing Tickets. Full unmatched repayment is external.
TC-22 BR-P09-06	Run PA4 mock checkout and PA5 Sandbox mode on isolated test data; inspect screens, records and summaries.	Mock results are explicitly simulated, not provider-verified; no real money. Sandbox mode verifies results server-side before eligibility/issuance. External readiness is not inferred from configuration alone.
TC-23 BR-P10-01, BR-P10-02	Cancel Pending; then Confirmed with exactly 12 hours remaining, 11:59:59 remaining and any Checked In Ticket. Try cancelling only one seat of a multi-seat Booking.	Pending can be cancelled before confirmation. Confirmed cancellation at ≥12 hours with no Checked In Ticket is eligible; later/checked-in requests are rejected. Only whole-Booking cancellation is supported.
TC-24 BR-P10-03, BR-P10-05	Cancel an eligible Confirmed Booking with paid fare 300,000 VND; cancel an unpaid Pending Booking.	Paid case records one Refund Pending for 270,000 and a 30,000 cancellation charge, without claiming cash repayment. Unpaid case has Refund Not Required. Statuses remain Not Required/Pending/Completed/Failed; external results are test fixtures, not in-app execution.
TC-25 BR-P10-03, BR-P10-07	Eligible paid fare is 100,001 VND; calculate a 90% refund.	For paid whole-Booking fare 100,001 VND, refund = 90,001 and cancellation charge = 10,000. For 100,005 VND, exact refund 90,004.5 rounds up to 90,005 and charge = 10,000. Calculate once on the Booking total, not per seat; 100-percent refunds and full unmatched repayment remain exact paid amounts.
TC-26 BR-P10-04, BR-P10-06	Cancel one Booking's A→B allocation while another owns B→C on the same seat. Repeat request and inject failure while recording cancellation/refund. Also use operator cancellation.	Release only the target Booking's Segment; unrelated B→C remains valid. Cancellation and Refund are separate; retries create no duplicate refund and failures cannot leave released seats with valid Tickets. Operator cancellation follows S04 with full refund.
TC-27 BR-P11-01, BR-P11-02, BR-P11-04	Vietnamese request specifies valid Stops, date, time, budget and vehicle type; seed more than five matching Scheduled Trips with different boarding times.	Interpret and validate supported filters; retrieve actual records; return at most five sellable Trips, ascending boarding time. No generated Trip/price replaces database facts.
TC-28 BR-P11-01, BR-P11-03, BR-P11-04	Use ambiguous date/Stop, unsupported preference, nonexistent Trip and a request to reserve/pay/cancel; seed no matching records.	Ask clarification or explain unsupported/no-match result and offer ordinary search. No invented records, guaranteed seat availability, hold or modifying action is produced.
TC-29 BR-P11-05, BR-P11-06	Use the configured Gemini Developer API Free Tier project/model; exceed five requests/minute per identity/session and the shared 200/day cap; force service delay beyond ten seconds.	Configured approved provider is used; sixth per-minute/excess daily request is limited. P11/P12 share cap. Fallback occurs by ten seconds; ordinary search remains available. Provider feasibility/model selection is evidence-gated.
TC-30 BR-P11-07	Submit input containing email/telephone/name; inspect payload construction, outbound requests and application database after the interaction.	PII fields and whole Booking records are not forwarded; no persistent conversation history is created in the business database. Do not claim anything about provider retention without separate

Case / references	Preconditions and action	Expected outcome
		evaluation.
TC-31 BR-P12-01, BR-P12-05	Anonymous asks FAQ/identified Trip facts; Passenger asks about own and another person's Booking; Staff/Admin request operational support; ask AI to modify a journey.	Public supported facts remain public. Private Booking answers require Passenger session plus ownership before retrieval/disclosure. Wrong-owner/role/action requests disclose nothing private and modify no records.
TC-32 BR-P12-02, BR-P12-03	Versioned demo FAQ covers booking, holds, payment, cancellation/refunds and check-in. Ask known/unknown questions and force FAQ retrieval failure.	Known answers cite the approved topic. Unknown answers use configured official contact, not invented policy/contact. No FAQ editor/live handoff is implied; unavailable content is an error, not a fabricated answer.
TC-33 BR-P12-04, BR-P12-06	Ask facts about an identified Trip versus asking for trip recommendations; exercise the same limit/timeout conditions as TC-29.	P12 supplies factual Trip/owned Booking support; P11 performs recommendations. Shared quotas and ten-second fallback apply; no persistent conversation history. Model/provider data handling remains evaluated in PA3.
TC-34 BR-S01-01, BR-S01-02, BR-S01-03	With Vietnam-local today, seed Trips and Tickets in all approved states. Request today, a seven-day range and an eight-day range, with Route/status filters.	Default is today; up to seven days is accepted and eight rejected. Display required Trip/vehicle/state fields; confirmed-issued count = Valid + Checked In, excluding Cancelled. Results sort by departure, read-only.
TC-35 BR-S02-01	Look up by Booking code, Ticket code and telephone shared by two Bookings.	Each supported key locates permitted records. Multiple matches stay separate until Staff explicitly selects; no arbitrary selection or check-in occurs.
TC-36 BR-S02-02, BR-S02-03, BR-S02-04	Inspect lookup fields, leave the screen, log out and expire a 15-minute Staff session.	Only operational names/contact telephone/journey/seat/Booking-Ticket-Payment states are visible; no passwords, auth data or provider technical details. Clear protected display on leave/logout/expiry; lookup remains read-only.
TC-37 BR-S03-01, BR-S03-02, BR-S03-03	Booking has two Valid Tickets. Staff selects one correct Trip/Ticket, reviews recorded boarding Stop and explicitly confirms.	S03 invokes C01, validates eligibility before confirmation and revalidates before save. Exactly selected Ticket becomes Checked In with Ticket/Booking/Trip IDs, Staff and time. Other Ticket is unchanged.
TC-38 BR-S03-02, BR-S03-04, BR-S03-05	Try unpaid/cancelled/wrong-Trip/already Checked In Tickets, Completed/Cancelled Trips, Departed first-Stop versus intermediate boarding, and reversal request.	Reject every invalid state or Stop; Departed intermediate boarding is permitted when Staff confirms the recorded Stop. No duplicate check-in, Payment mutation, Ticket issuance or seat release. Reversal is unsupported.
TC-39 BR-S03-03, BR-S03-04	Staff declines confirmation; then force save failure, concurrent double confirmation and session expiry between review and save.	Decline changes nothing. Pre-save C01 rejects expired/changed eligibility; no partial successful check-in or duplicate audit success survives failure/concurrency.
TC-40 BR-S04-01, BR-S04-02, BR-S04-03	Attempt each transition, including departure before/at first-Stop planned time, missing vehicle, cancellation without reason, rollback and terminal-state changes.	Only Scheduled→Departed→Completed or Scheduled→Cancelled is allowed with stated conditions. Cancellation needs reason; departure needs valid assignment and time reached. Save Staff/time/old-new status/reason; reject invalid

Case / references	Preconditions and action	Expected outcome
		transitions without mutation.
TC-41 BR-S04-04, BR-S04-05	Cancel a Scheduled Trip containing Pending/Confirmed Bookings and Valid/Checked In Tickets; replay cancellation with an existing refund request.	Invalidate all affected Tickets, cancel affected Bookings, release allocations and prevent check-in. Eligible paid fare gets a single 100% refund request; reconcile existing requests so repayment never exceeds payment.
TC-42 BR-S04-06	After operator cancellation, deliver a late payment success; inspect displayed cancellation/refund results.	No Ticket is issued or revived. Payment follows P09 reconciliation/full unmatched repayment; cancellation success never claims external refund execution completed.
TC-43 BR-A01-01, BR-A01-02	Create each of the two preset vehicle types. Try duplicate code/plate/seat ID, missing required fields and capacity different from seat count.	Only a complete unique vehicle with preset seated/sleeper layout and matching capacity is saved. No custom layout/type editor is required; invalid input changes nothing.
TC-44 BR-A01-03, BR-A01-04	Try structural/capacity edit and deletion after assignment to a Trip; separately deactivate a used vehicle and delete a never-used one.	Used structure/dependent history is protected; deactivation prevents new assignment without invalidating current/sold Trips. Only unused unreferenced vehicle can be deleted after confirmation.
TC-45 BR-A02-01, BR-A02-04	Create Route with two distinct Stops and valid names/addresses; test one Stop, repeated Stop, duplicate code and reversed Segment.	Require unique Route and ordered distinct Stops. Invalid structure/Segment is rejected. Availability and consecutive-leg fare use that saved Stop order.
TC-46 BR-A02-02, BR-A02-03	Edit Stop order or delete Route/Stop after use; attempt equivalent rerouting with a new Route and retire the old one.	Used structure/references remain immutable; rerouting creates a new Route identity. Delete only unused unreferenced records; retirement blocks new assignments while preserving history.
TC-47 BR-A03-01, BR-A03-02	Prepare a 30-day generation with Route, operating days, assigned vehicle, increasing Stop times and complete nonnegative leg fares. Preview, decline, then confirm; try 31 days and missing/negative fares.	Nothing generated before confirmation. Valid ≤30-day preview creates complete concrete Trips; invalid input cannot be confirmed. Segment fare is leg sum, identical across seats. Zero-fare integration feasibility remains an explicit PA3 check, not proof of provider support.
TC-48 BR-A03-03	Generate overlapping vehicle intervals, unordered Stop times and the same schedule/date range twice.	Reject overlapping Trip intervals, invalid times and duplicate Trip creation; a validation/save failure creates no partial set of Trips.
TC-49 BR-A03-04	Preview an edit, create a live hold before save, then try to change vehicle/Route/time/fare. Repeat with a Confirmed Booking and with expired historical Bookings.	Recheck at save; live/confirmed allocations block structural/time/fare changes. No sold/held facts or historical Booking/Ticket journey/fare snapshots are overwritten.
TC-50 BR-A03-05, BR-A03-06	Change a schedule and regenerate later dates; attempt deletion of referenced schedule/Trip.	Existing concrete Trips do not change retroactively. Delete only unreferenced records; stop future generation or use Scheduled-Trip cancellation while preserving history.
TC-51 BR-A04-01,	Administrator opens overview, uses each approved filter and selects a Booking;	Only read/search/filter is allowed. Necessary contact/journey/lifecycle details may be shown;

Case / references	Preconditions and action	Expected outcome
BR-A04-02, BR-A04-06	attempt seat edit, cancellation, manual payment approval and refund execution.	passwords/auth data are excluded. Modifying/overriding actions are unavailable/denied and never bypass P09 eligibility.
TC-52 BR-A04-03, BR-A04-04, BR-A04-05	Seed known Booking/Ticket counts, verified payments, an extra successful transaction and Review Required reconciliation, plus labelled mock records.	Count statuses exactly; confirmed-issued = Valid + Checked In. Show successful money recorded, not revenue/profit; retain separate reconciliation/refund states and flag simulated figures, without treating mock as verified money.
TC-53 BR-C01-01, BR-C01-02	Invoke C01 with authorized and unauthorized session, missing Ticket, wrong Trip/Stop and current valid records; compare pre/post journey records.	Return Eligible or explicit mapped failure; never mutate journey state. First invocation returns recorded Stop for Staff review; pre-save invocation checks reviewed Stop and current eligibility. S03 performs the write, not C01.
TC-N01 NFR01	Run booking and Staff check-in at 360/390/430 px and desktop 1280 px; use latest two stable Chrome/Edge/Safari releases available at execution.	No clipped required controls/horizontal page overflow; each error/success reference matches the UC. Record exact versions/devices, viewport and screenshots; this is a test design, not a completed browser test.
TC-N02 NFR02	For each P04 and P05 use 20 Routes/100 Stops/20 Vehicles/500 Trips/5,000 Bookings including holds/expired/cancelled; 20 concurrent sessions; ten warm-ups then ≥100 measured actions per UC.	Record client/server/network and submit/open-to-stable-render measurements. Each UC p95 ≤3 seconds, including app/DB/render time; external AI/payment latency is excluded and tested separately. Retain samples and percentile calculation.
TC-N03 NFR03	Run TC-13/14/17 concurrently and at expiry boundaries; observe committed allocations after each operation/failure/retry.	Zero committed effective overlaps for one Trip/seat. Adjacent Segments can reuse it. Expired holds cannot authorize confirmation; retain allocation-state and timing evidence.
TC-N04 NFR04	Exercise every access-matrix deny cell, including altered Booking/Ticket IDs and revoked/expired sessions; inspect logs/source/config for secrets.	No denied action exposes protected data or changes records. No secrets in tracked source/log output. API and UI checks agree; retain per-cell results and observed response rather than assuming UI hiding is enforcement.
TC-N05 NFR05	Execute TC-18–22 with signed Sandbox fixture callbacks, duplicates, wrong identity/amount, late/extra outcomes and client-only redirects.	Only authoritative eligible success issues Tickets exactly once; no invalid/late result issues one. Preserve separate attempt/reconciliation states and callback evidence.
TC-N06 NFR06	Inject retrieval/write/network failure at hold creation, payment confirmation, cancellation/refund recording and check-in; retry and refresh each.	No partial effective allocation, duplicate Ticket/refund/check-in, or false success survives. Capture before/after states and recovery result for each failure point.
TC-N07 NFR07	Review an actual PR and endpoint inventory once implemented; link each critical BR test to its changed component/API.	At least one other member review before merge; retained automated test results for the specified critical scenarios. Every endpoint records inputs/outputs/permissions/errors, defects have owner/status. No implementation evidence is fabricated at design time.
TC-N08	Use versioned Vietnamese set: valid request,	Every factual answer in this finite acceptance set is

Case / references	Preconditions and action	Expected outcome
NFR08	ambiguous Stops/date, unsupported preference, no Trip, unknown policy, wrong-owner Booking, modifying request, provider failure and timeout; compare all displayed facts with fixture records/FAQ.	grounded and authorized; unsupported cases use specified fallback, and no modifying action is performed. Record exact inputs/retrieved facts/outputs/model/config; this does not assert universal AI correctness.
TC-N09 NFR09	On a clean classroom environment, follow setup/configuration/data-load instructions and run public search plus role-protected smoke paths; separately connect AI/Sandbox if credentials are available.	Reproducible setup and known test data. Readiness marked verified only with actual connection evidence; missing credentials remain blocked and simulated modes are labelled. No production availability/payment certification claim.

D.5. Approved decisions

ID / topic	Approved rule and example	Impact / state
DEC-01 — Travel date for intermediate boarding	Approved option A — Passenger search uses the selected boarding Stop's calendar date in Asia/Ho_Chi_Minh. Example: A departs 10 Nov 23:00; B boards 11 Nov 01:00. An advance B→C search for 11 Nov matches while before the common first-Stop cutoff. First-Stop operational departure date remains separately labelled.	P04/P11 filter semantics, UI labels, TripStop query and overnight test TC-09. Does not change common first-Stop booking cutoff. Approved A by Liên Trung Hiếu, 4 November 2026 (user confirmation: 1A 2A).
DEC-02 — Integer-VND refund rounding	Approved option A — Round 90 percent of the whole-Booking paid fare to nearest integer VND, halves up; charge = paid fare minus rounded refund. Example 100,001 VND: refund 90,001; charge 10,000. Apply floor((9F+5)/10) for integer nonnegative F; never sum individually rounded per-seat refunds.	P10 refund amount and stored financial snapshot; operator 100% and full unmatched repayment remain exact paid amounts. TC-25 expected outcomes are finalized; execution evidence is not yet available. Approved A by Liên Trung Hiếu, 4 November 2026 (user confirmation: 1A 2A).

D.6. Source and validation register

AS-01 — Single operator and manual phone/chat/spreadsheet baseline are classroom scenario assumptions, not field-survey findings. The project lead approved the PA2 correction scope on 4 November; no real operator interview or individual member sign-off is claimed. Existing approved 10-minute hold, first-Stop cutoff, whole-Booking cancellation, refund percentages and per-Ticket check-in remain unchanged.
AS-02 — AI/VNPay credentials, provider-side data handling and deployment feasibility require actual PA3 evidence. Test-case coverage does not prove an implementation passes; candidate API/schema/interaction targets must be reviewed against each linked BR before coding.

D.7. UML model reading rules

The source contains six views: overview, Passenger discovery, booking/payment/cancellation, AI assistance, operations and shared authentication. Repeated actor symbols are references to the same role. System User generalization shares login/logout with Passenger, Staff and Administrator. Supporting C01 factors existing eligibility checks; it is not a 21st user goal. The overview suppresses some detail while the booking view shows EP-P06-CANCEL inside P06.
The extend condition cancelRequested means Passenger chooses Cancel from the owned Booking details shown by P06. P06 is complete without cancellation; P10 applies its own eligibility/confirmation checks and also supports direct authorized entry. S03 includes C01 before review and again before save. P08 creates Pending + fixed

ten-minute hold and does not include P09. AI/VNPay are supporting external services; explanations and scope text are kept in this document, not as diagram notes.

Layout baseline 1.5 — Approved by the project lead on 4 November 2026. Actor associations use aligned vertical segments with horizontal branches. Every branch remains a separate actor-to-UC association in the editable source; shared geometry creates no new UML element or UC-to-UC relationship. Includes, extends and generalizations retain their own notation and direction. This is a layout-only amendment.

D.8. PA3 integration revision — 6 November 2026
Edition 1.8 inherits baseline 1.5 and editions 1.6/1.7, with a leader-approved provider amendment on 8 November 2026 (QĐ-07). BR-P11-05/BR-P12-06 now name Google Gemini Developer API Free Tier; UC IDs, actor topology, 96 primary BR identifiers and 62 acceptance designs remain. SAD 1.2 records model/quota/data handling; software tests and provider connectivity remain unverified.
REF-01 — PA3_Group11_UseCaseModel_v2.1.drawio — editable six-view UML source.
REF-02 — PA3 UC Model v2.1 — six full-size PNG exports (uc-overview, uc-discovery, uc-booking, uc-ai, uc-operations, uc-auth); matching SVG exports. This set replaces the UC views in PA3 Diagrams v2.0.
Editable model contains overview plus discovery/booking/AI/operations/shared authentication. UML semantics remain standard; styling follows the common PA3 palette. Versions 1.5 and 2.0 remain historical; visual edition 2.1 is current.